

David L. Nelson and Michael M. Cox

LEHNINGER

PRINCIPLES OF BIOCHEMISTRY

Sixth Edition

CHAPTER 16

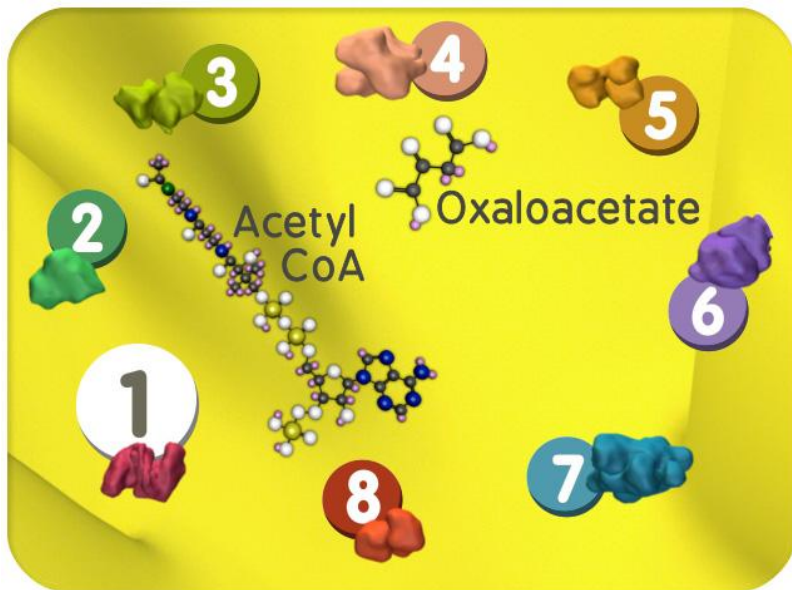
The Citric Acid Cycle



Today in this class...

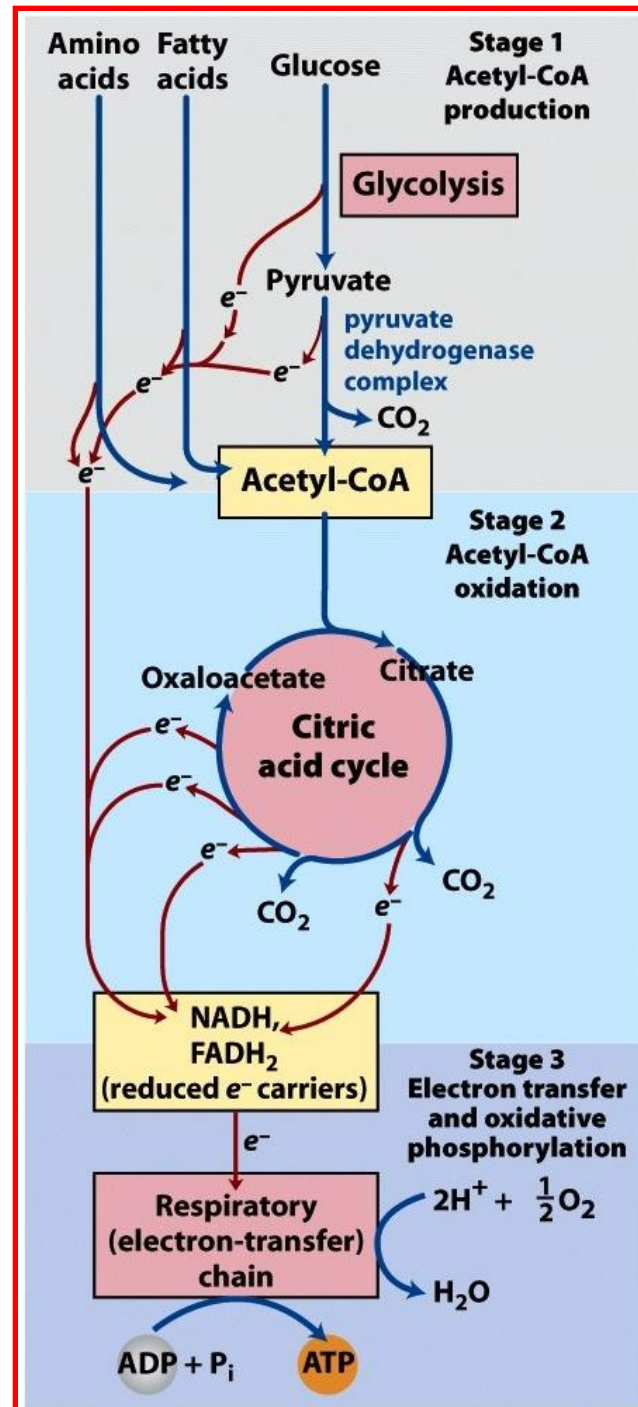
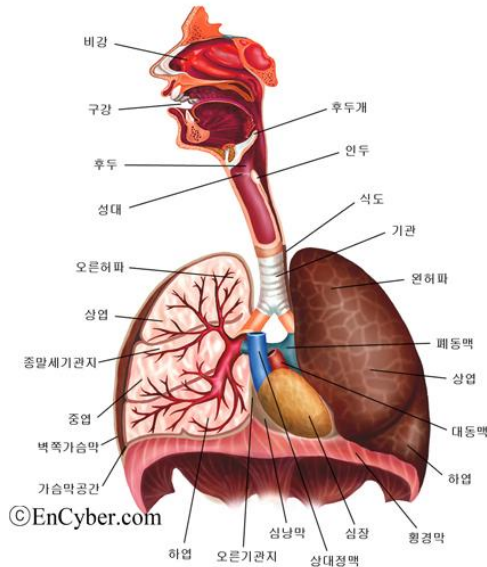
Citric acid cycle

Regulation of citric acid cycle



Stages of Cellular Respiration

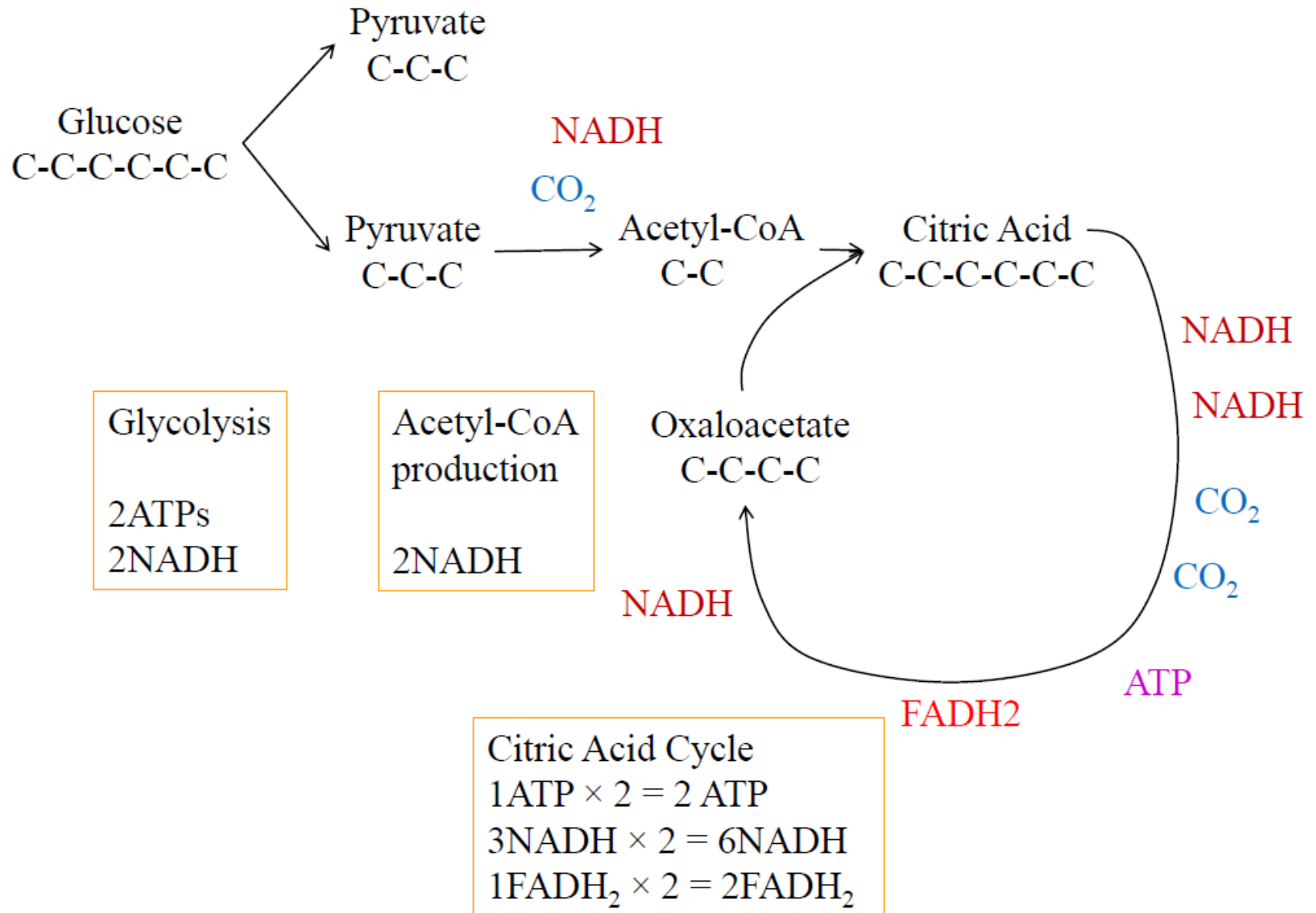
Respiration ?



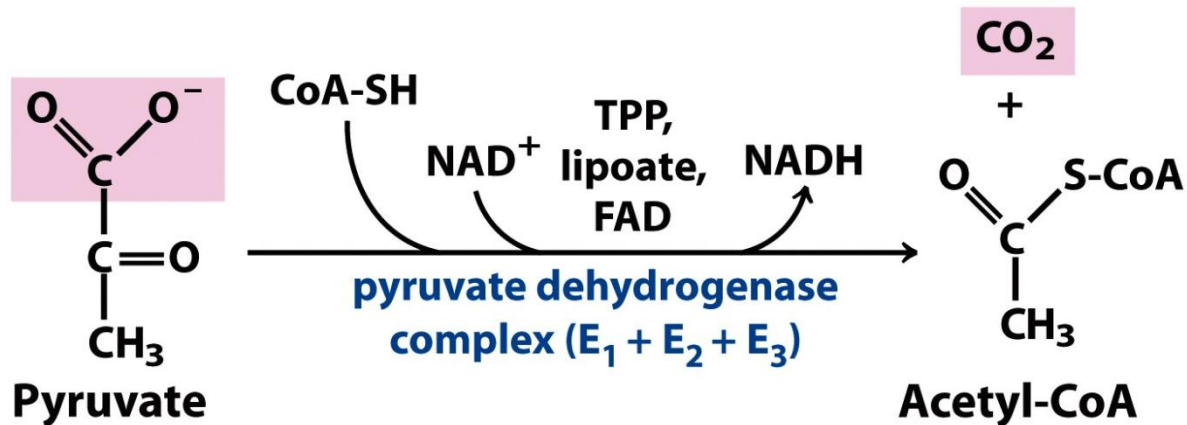
Respiration !!



Overview of citric acid cycle



16.1 Production of Acetyl-CoA

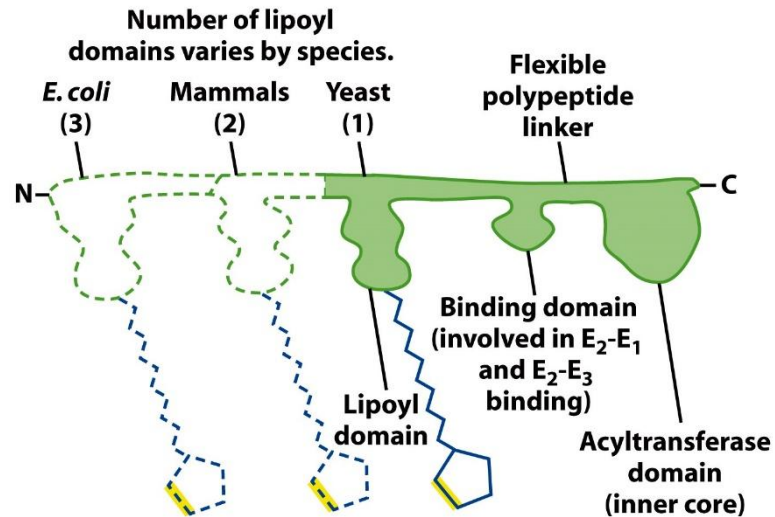
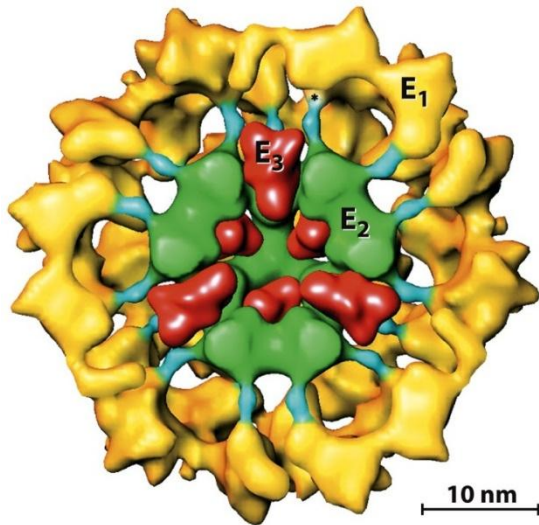


PDH complex consists of 3 enzymes

E1: pyruvate dehydrogenase (TPP)

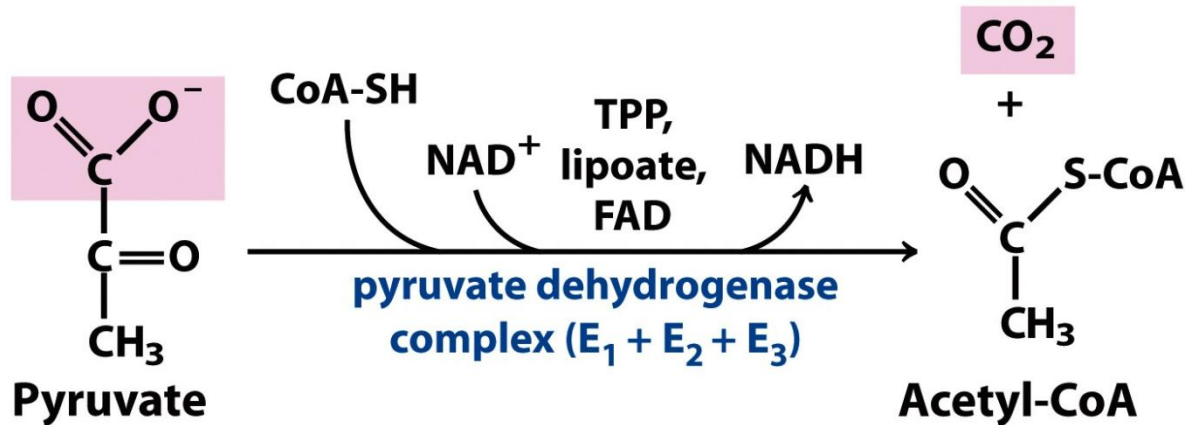
E2: dihydrolipoyl transacetylase (Lipoate, CoA)

E3: dihydrolipoyl dehydrogenase (FAD, NAD)



The long lipoyllysyl arm swings from E1 to E2 to E3, tethering the intermediates to allow substrate channeling

Pyruvate is oxidized to Acetyl-CoA and CO₂



- Oxidative decarboxylation & Irreversible reaction
- 5 coenzymes involved

CoA: coenzyme A (pantothenic acid; B₅) – acetyl group

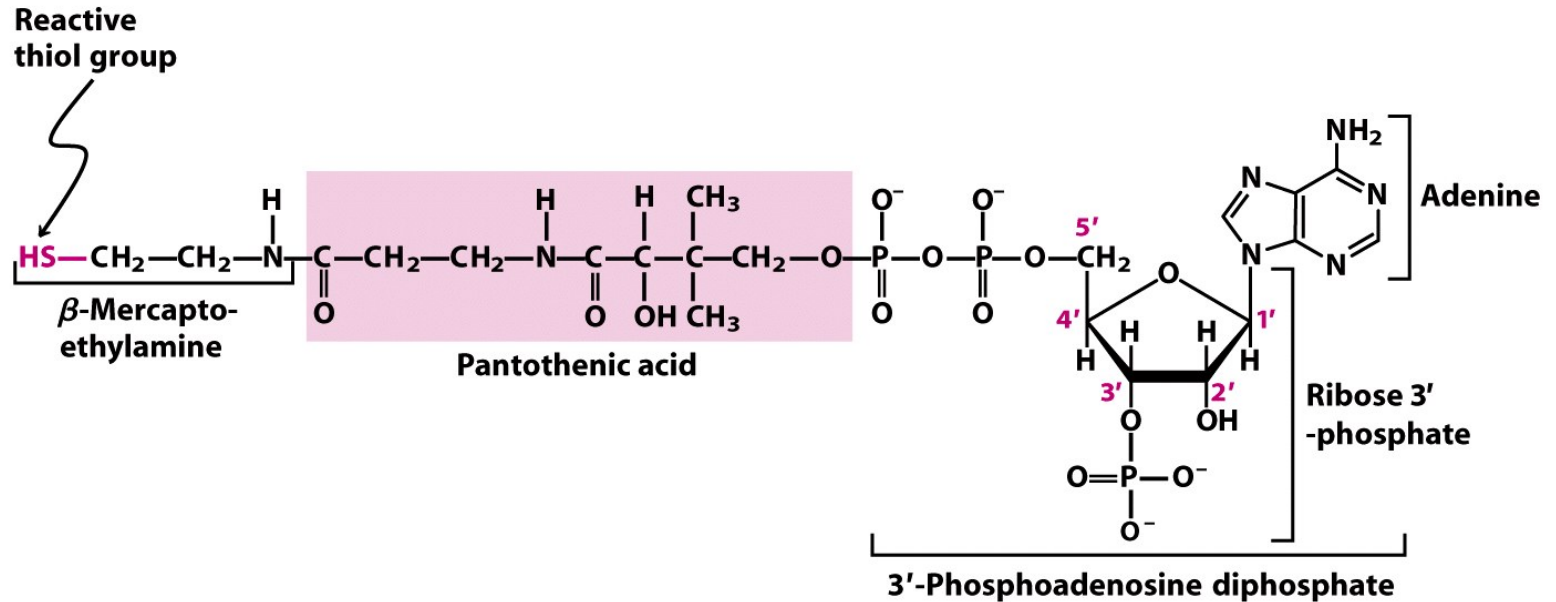
NAD: nicotinamide adenine dinucleotide (niacin; B₃) - electron

FAD: flavin adenine dinucleotide (riboflavin; B₂) - electron

TPP: thiamine pyrophosphate (thiamine; B₁) – hydroxyethyl group

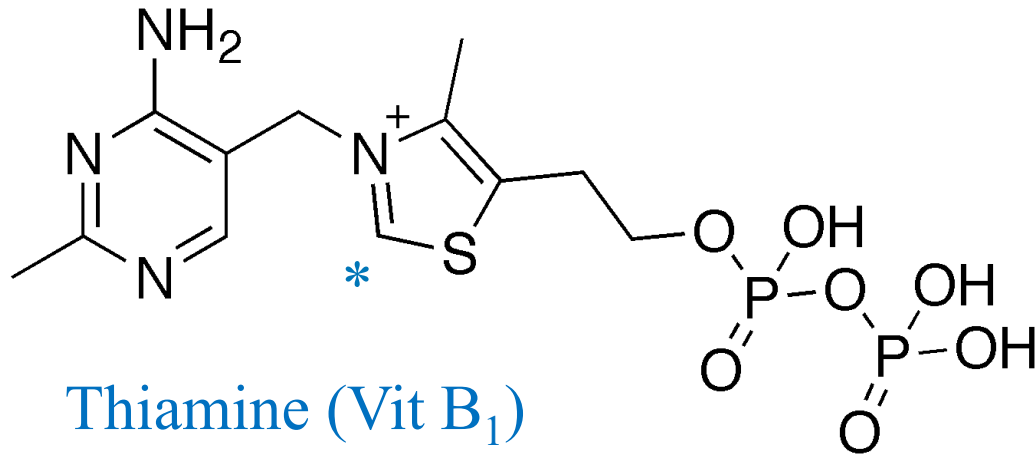
Lipoic acid (lipoate) – acetyl group

Coenzyme A – acetyl group (acyl)



Fatty acyl-CoA
Acetyl CoA

Thiamin pyrophosphate (TPP) - hydroxyethyl group (2C)



Pyruvate dehydrogenase complex (TCA)

Pyruvate decarboxylase (ethanol fermentation)

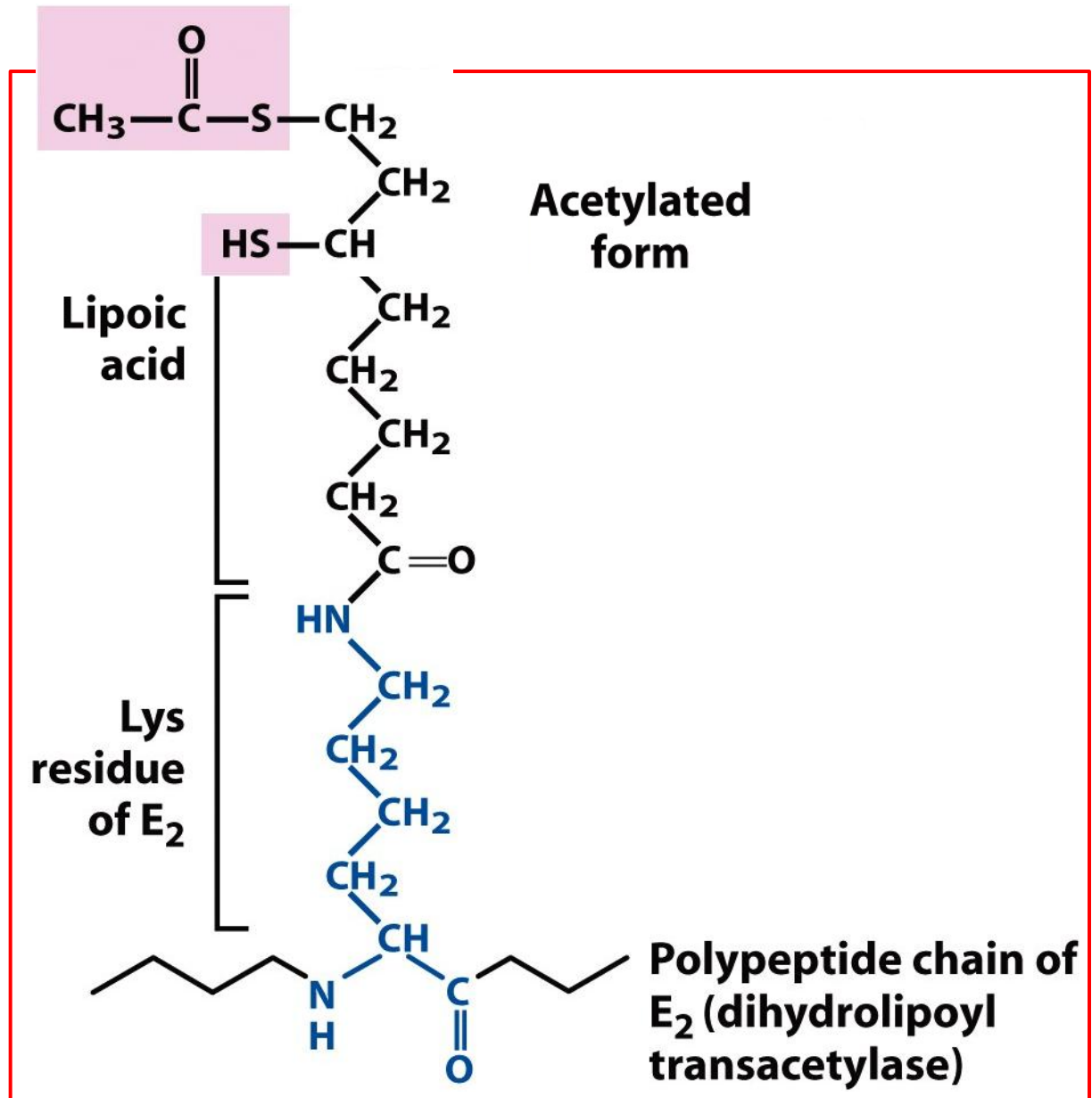
α -ketoglutarate dehydrogenase complex (TCA)

Transketolase (Pentose-phosphate pathway)

Branched-chain amino acid dehydrogenase complex

Lipoic acid (lipoate) – acetyl group

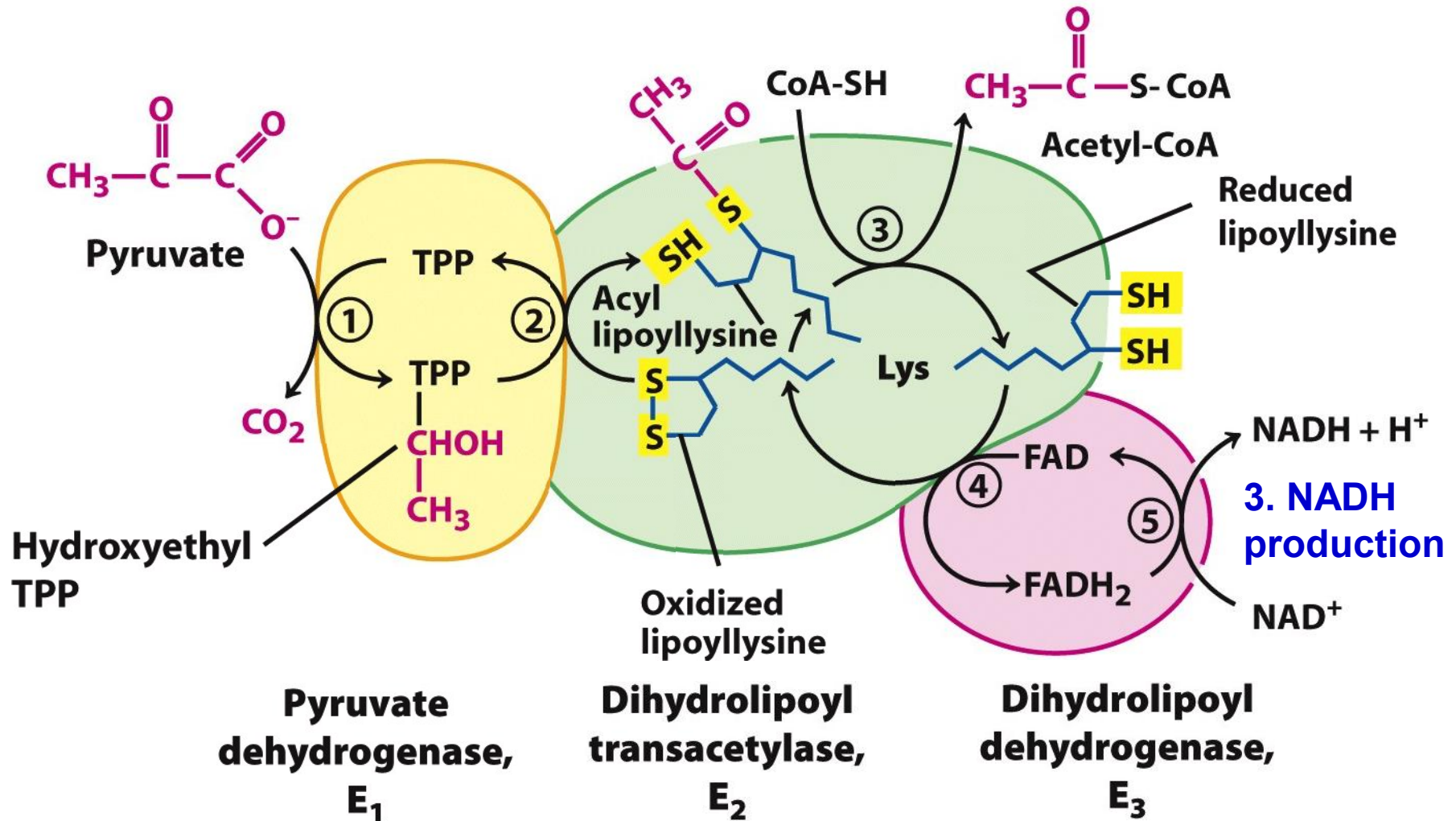
- Attached to Lys residue of E2
- Can be acetylated. (to transfer acetyl group)



Pyruvate dehydrogenase (PDH) reaction

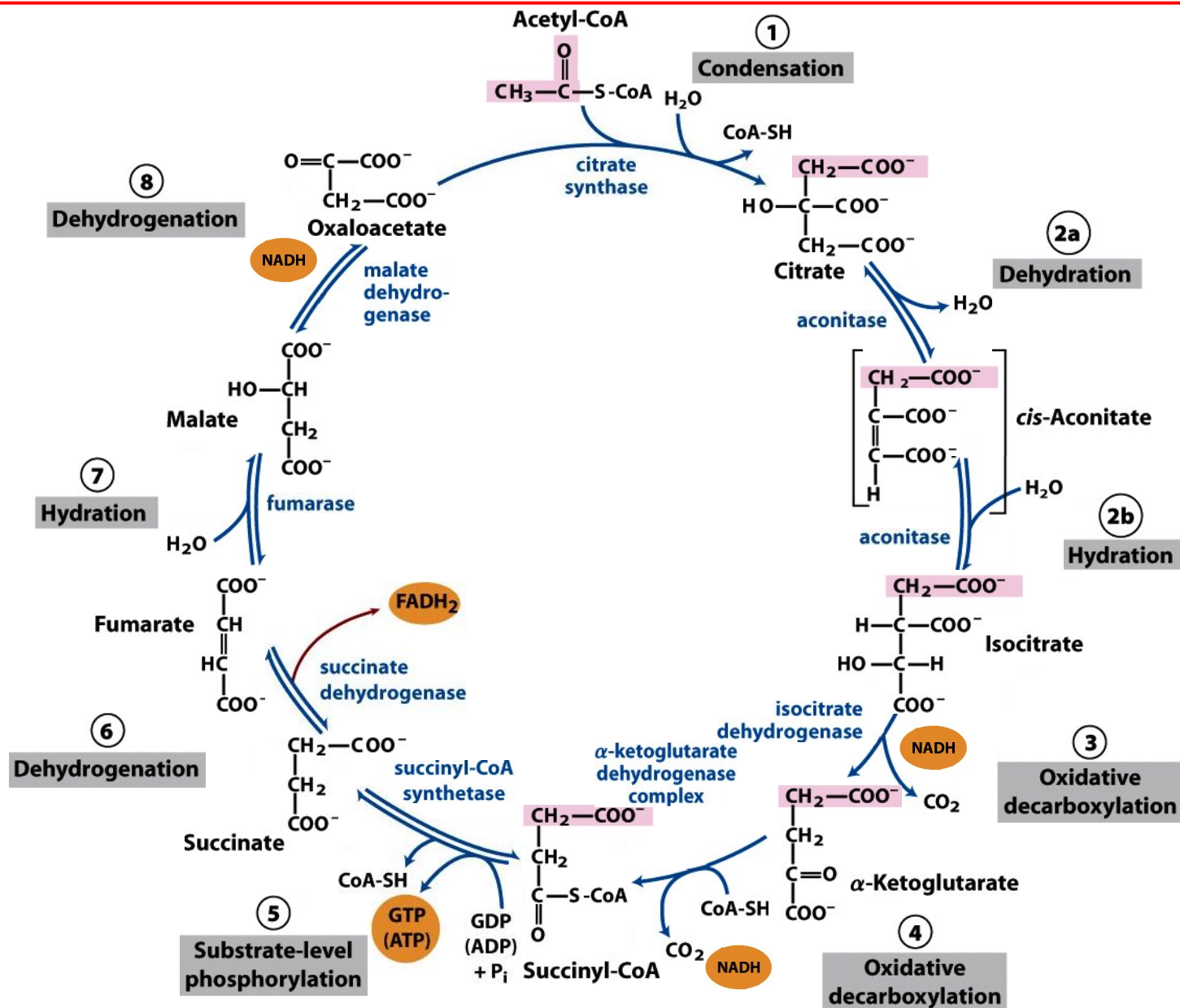
1. decarboxylation

2. acetyl-CoA production

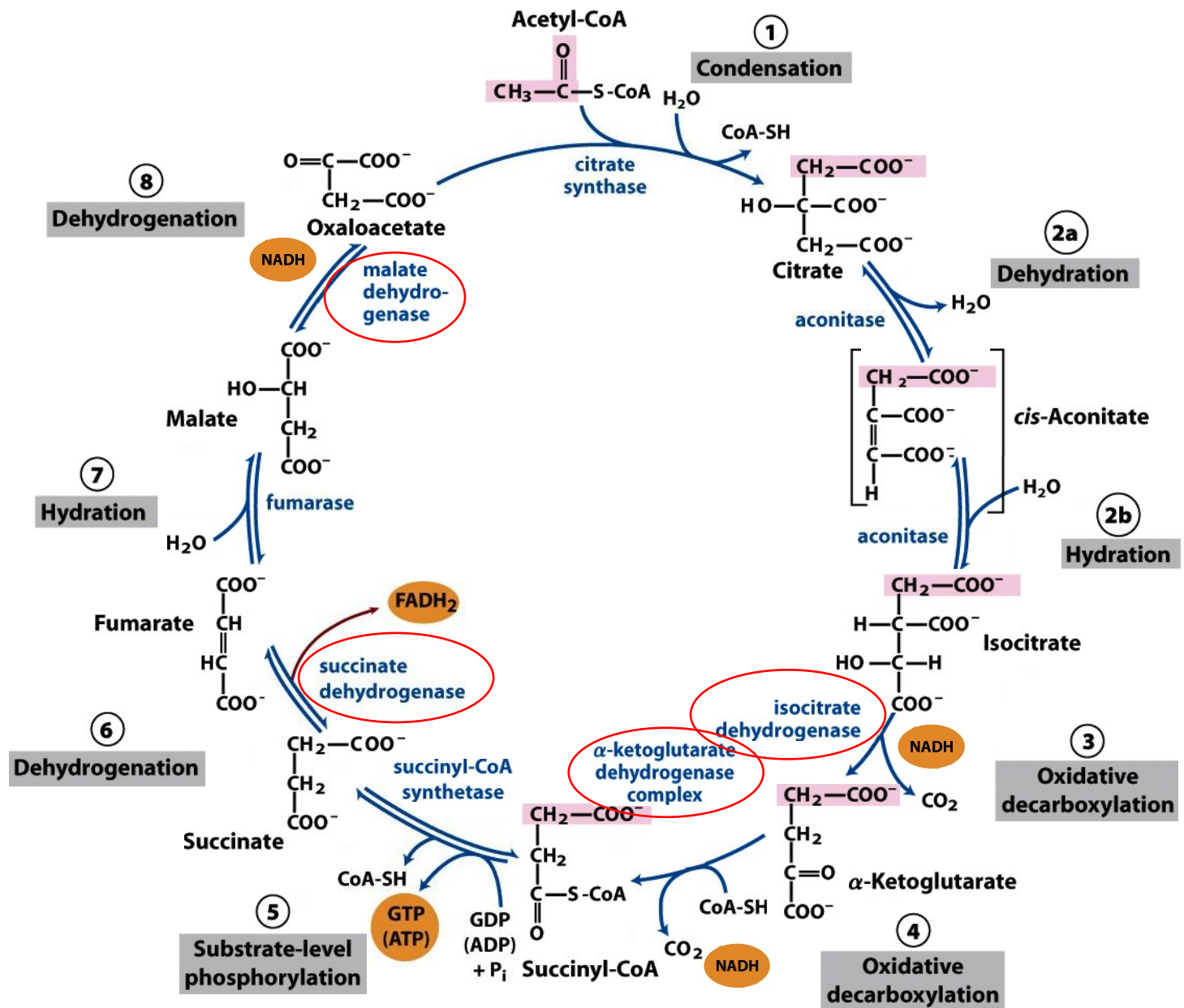


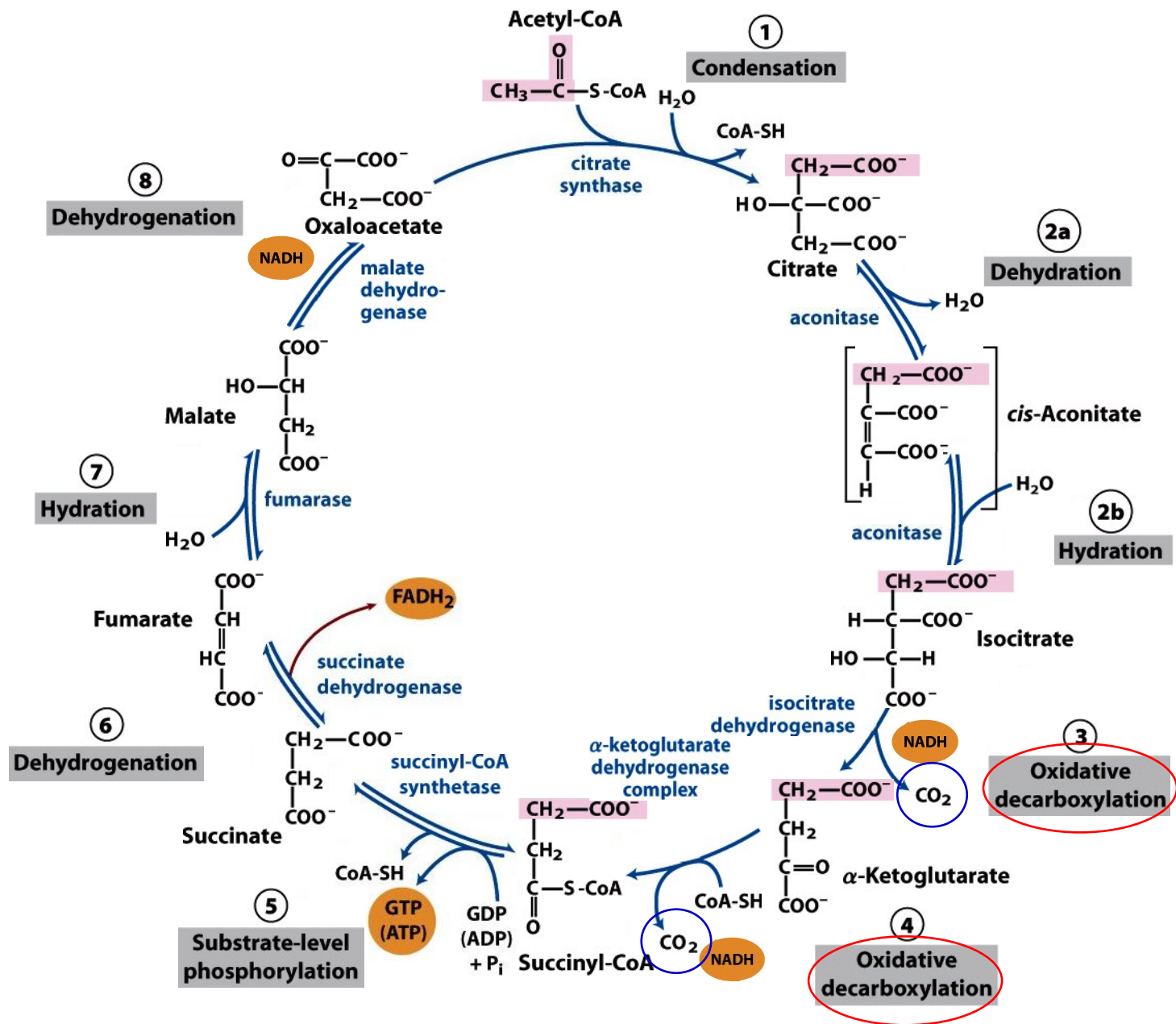
16.2 Reactions of the Citric Acid Cycle

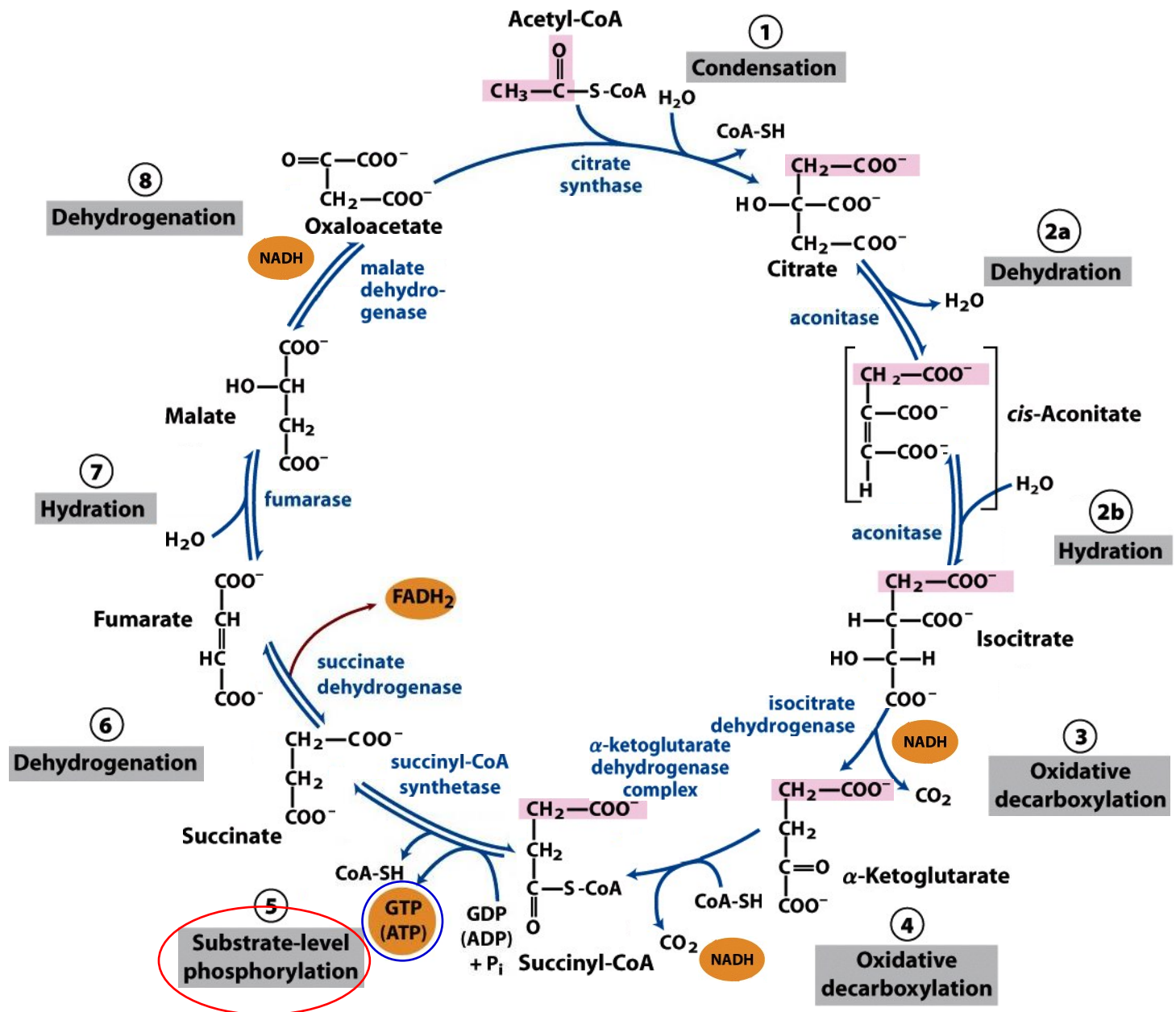
All reactions take place in mitochondria



All reactions take place in mitochondria

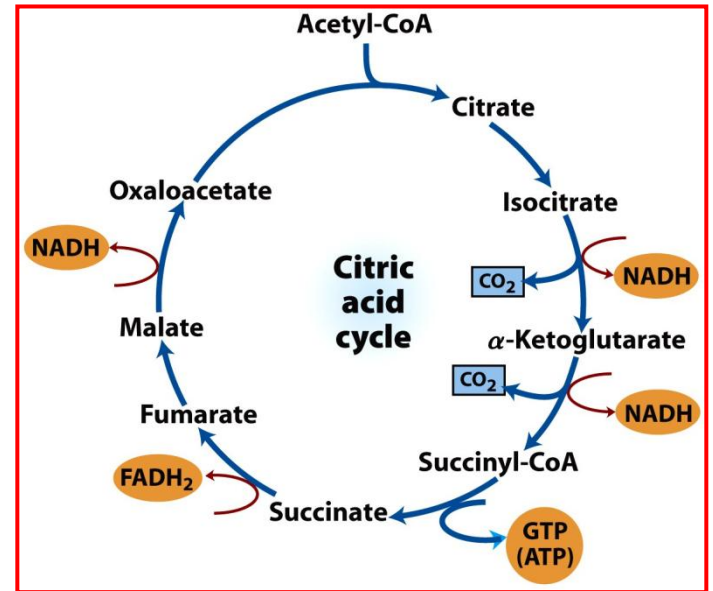


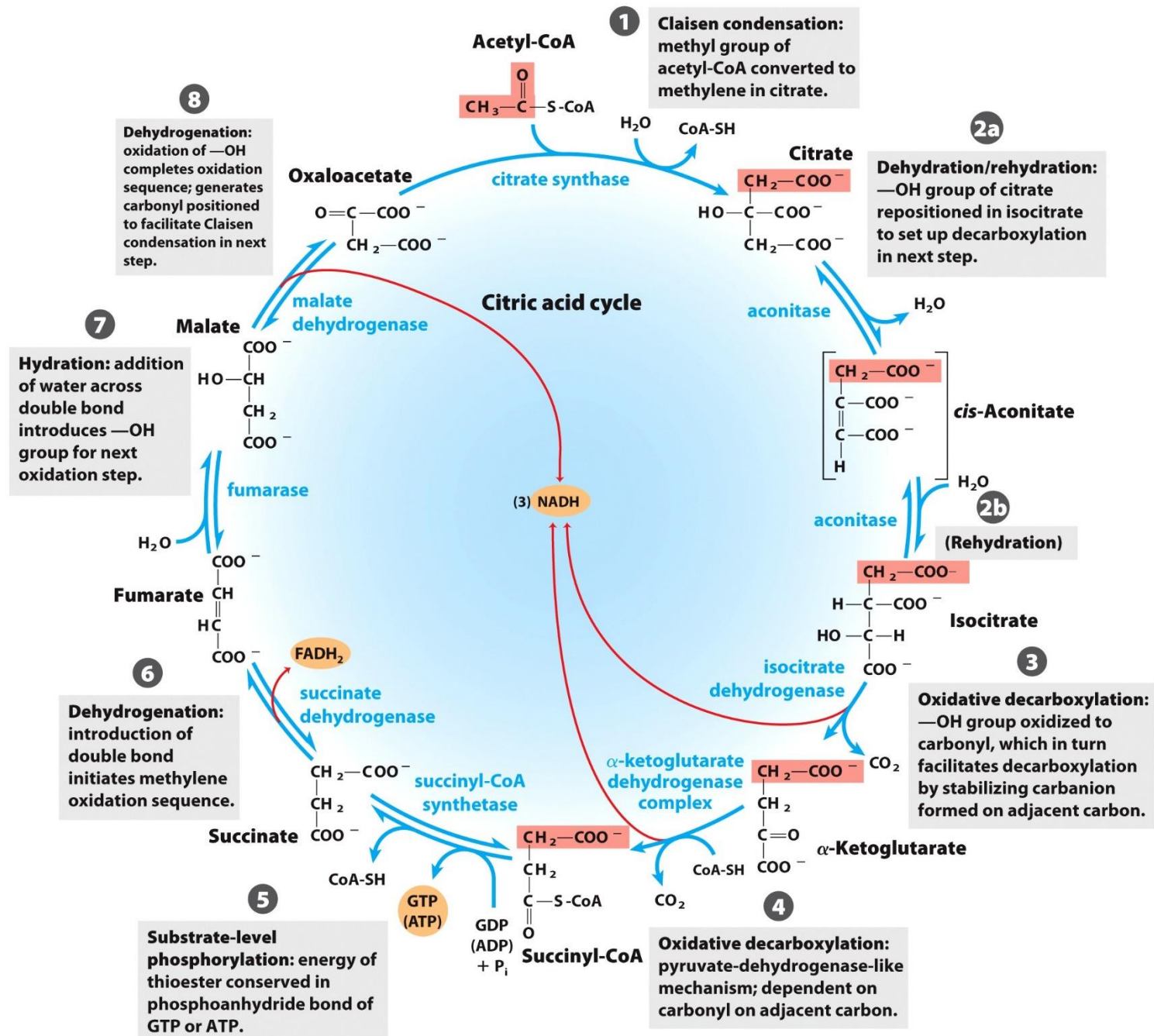




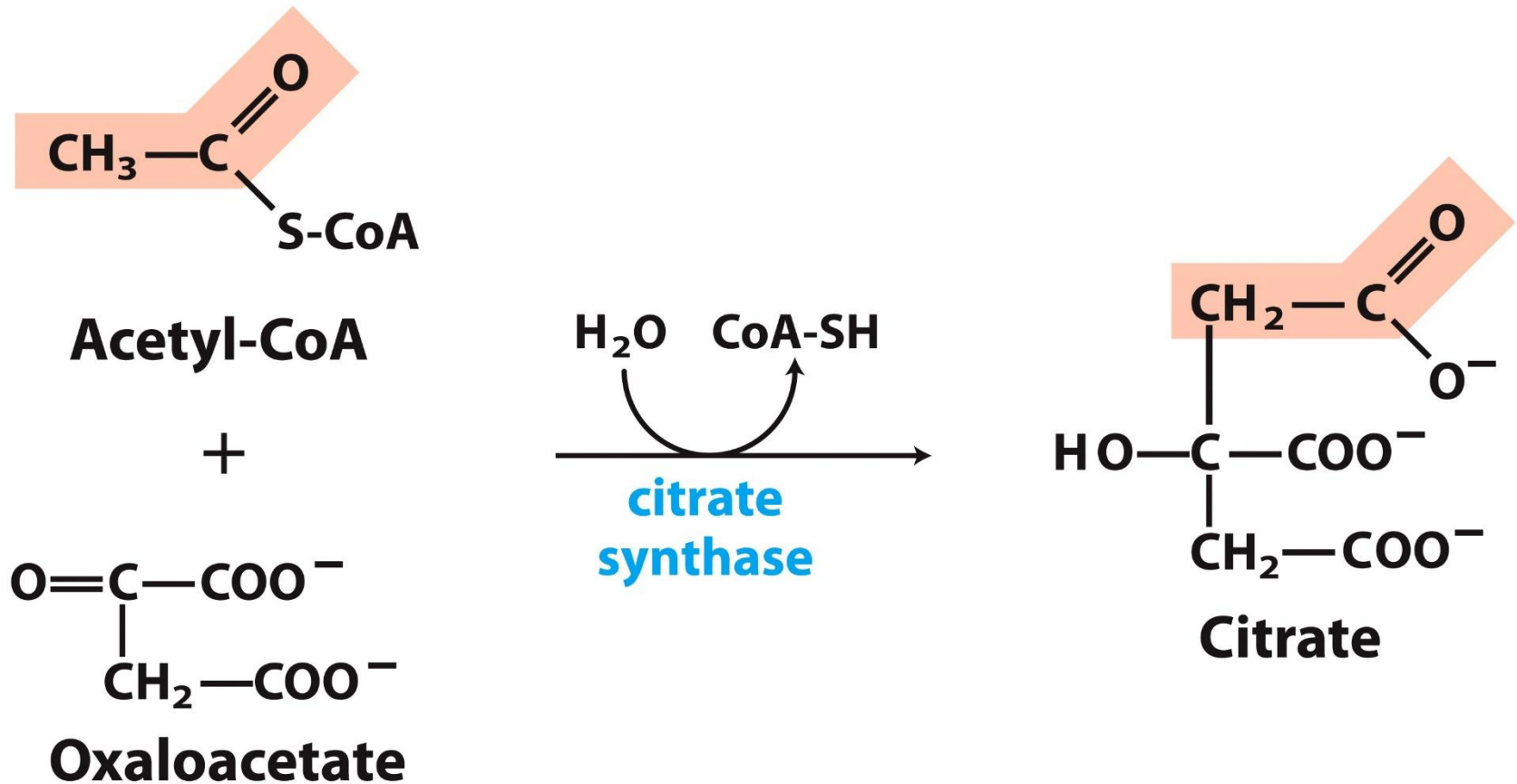
The overall outcome from citric acid cycle

- 2-C acetyl group enters
- 2 C leave as CO_2
- E production:
3 NADH (2.5 ATP/NADH)
1 FADH_2 (1.5 ATP/ FADH_2)
1 GTP (ATP)

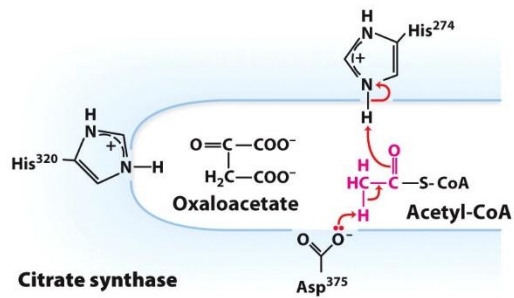




1) Claisen condensation : methyl group of acetyl-CoA converted to methylene in citrate

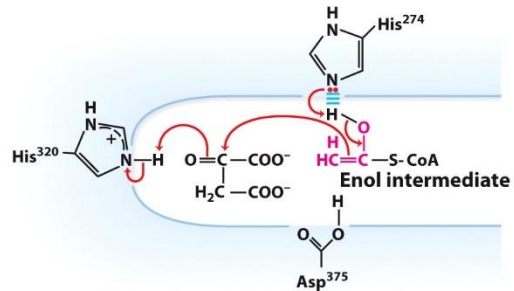


$$\Delta G'^{\circ} = -32.2 \text{ kJ/mol}$$



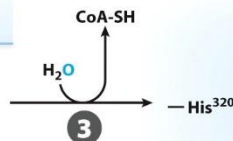
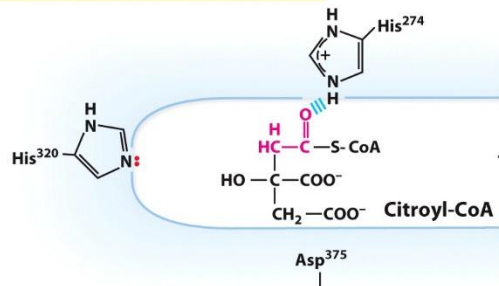
The thioester linkage in acetyl-CoA activates the methyl hydrogens. Asp³⁷⁵ abstracts a proton from the methyl group, forming an enolate intermediate. The intermediate is stabilized by hydrogen bonding to and/or protonation by His²⁷⁴ (full protonation is shown).

1



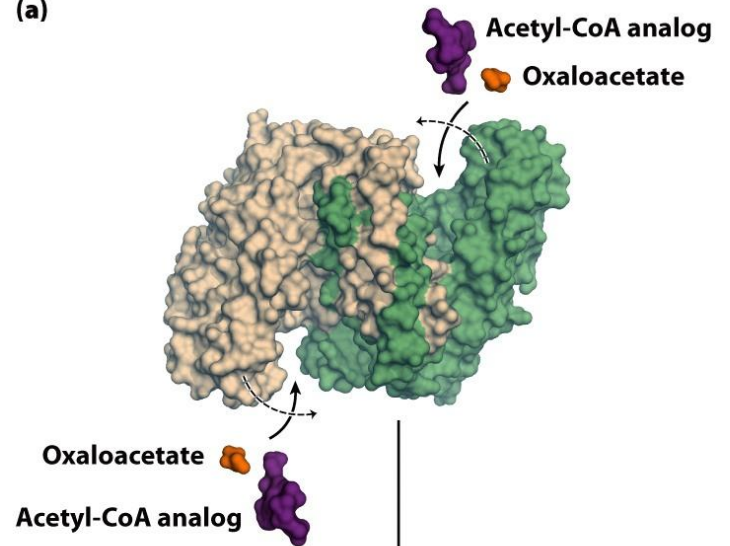
The enol(ate) rearranges to attack the carbonyl carbon of oxaloacetate, with His²⁷⁴ positioned to abstract the proton it had previously donated. His³²⁰ acts as a general acid. The resulting condensation generates citryl-CoA.

2

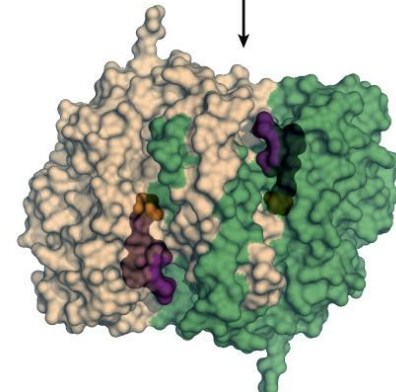


The thioester is subsequently hydrolyzed, regenerating CoA-SH and producing citrate.

(a)

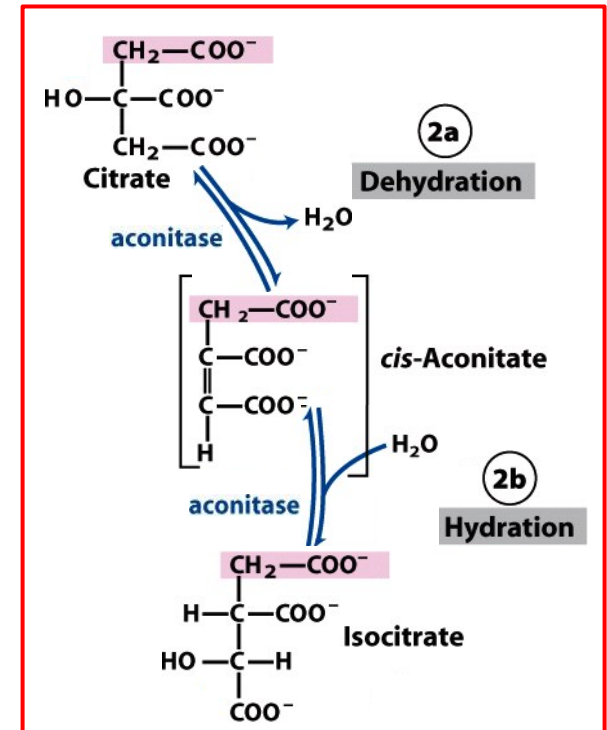
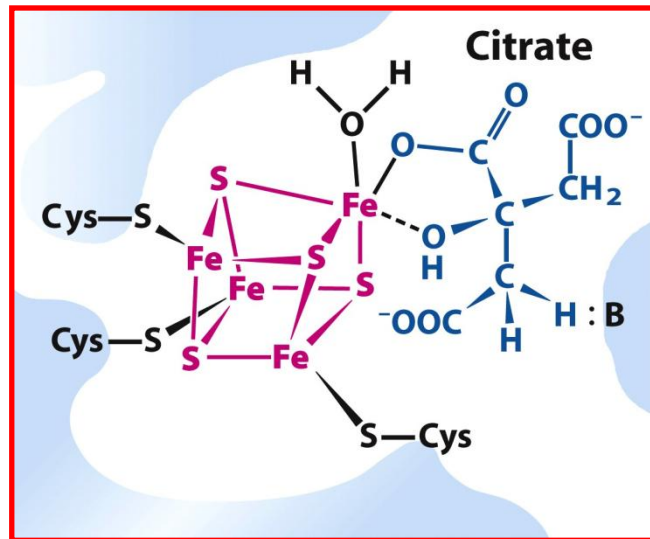


(b)

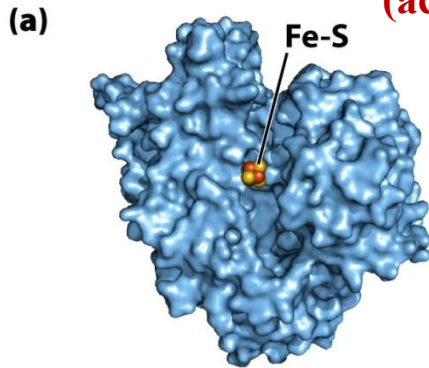


2) Formation of isocitrate via cis-aconitate

- Aconitase has an iron-sulfur center: **1. binding of Substrate,**
2. addition or removal of H₂O.

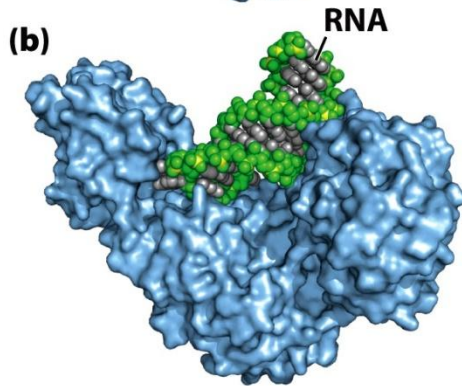


(aconitase: enzyme in citric acid cycle)

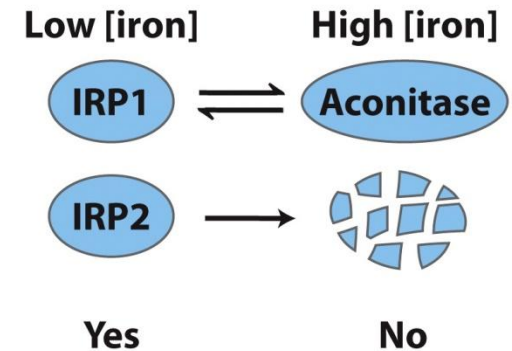


Aconitase is a moonlighting enzyme

(Working a second job, in particular one after normal business hours in the evening or night)



트랜스포머? 투잡?

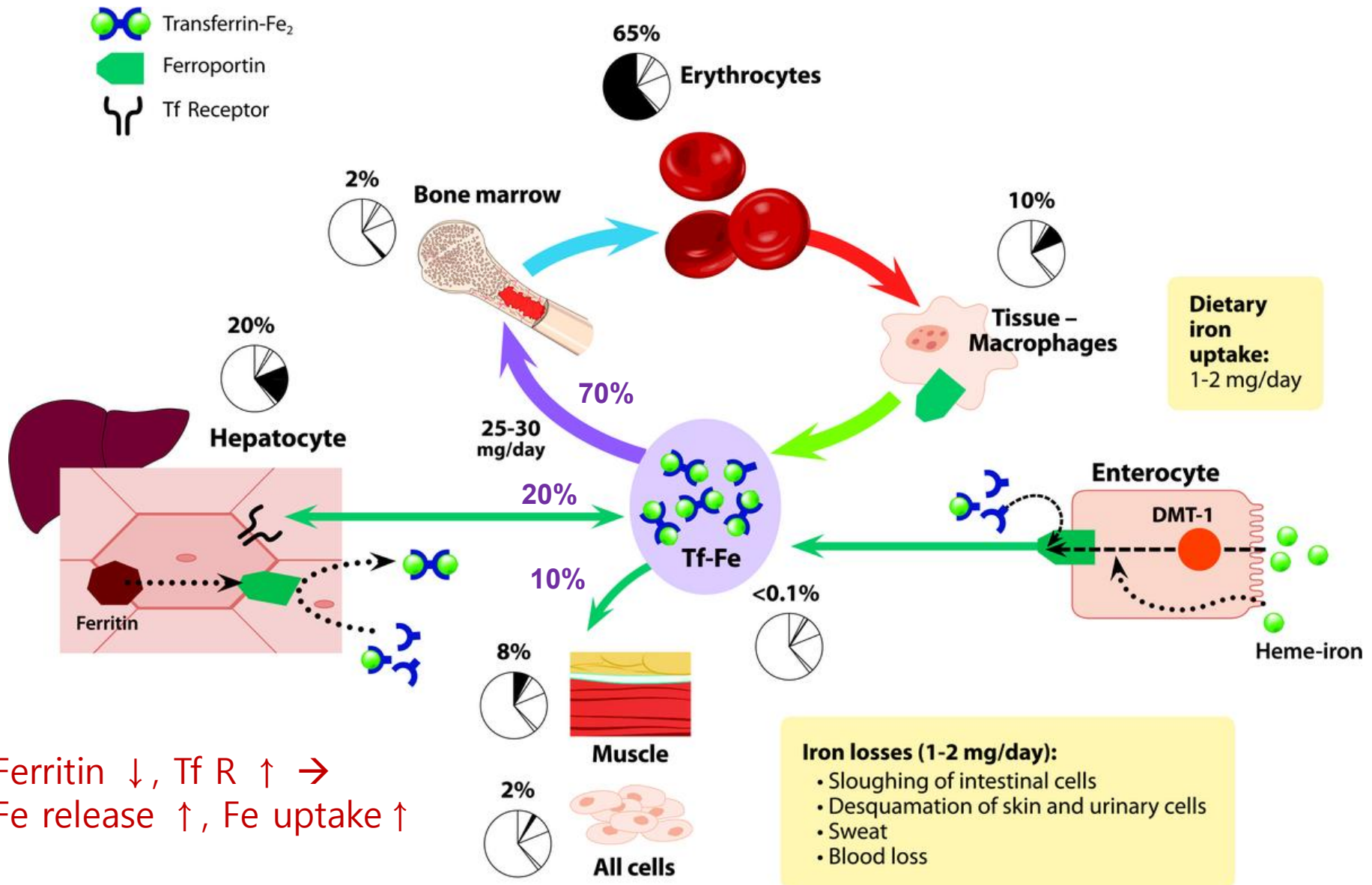


(IRP: iron response element binding protein)

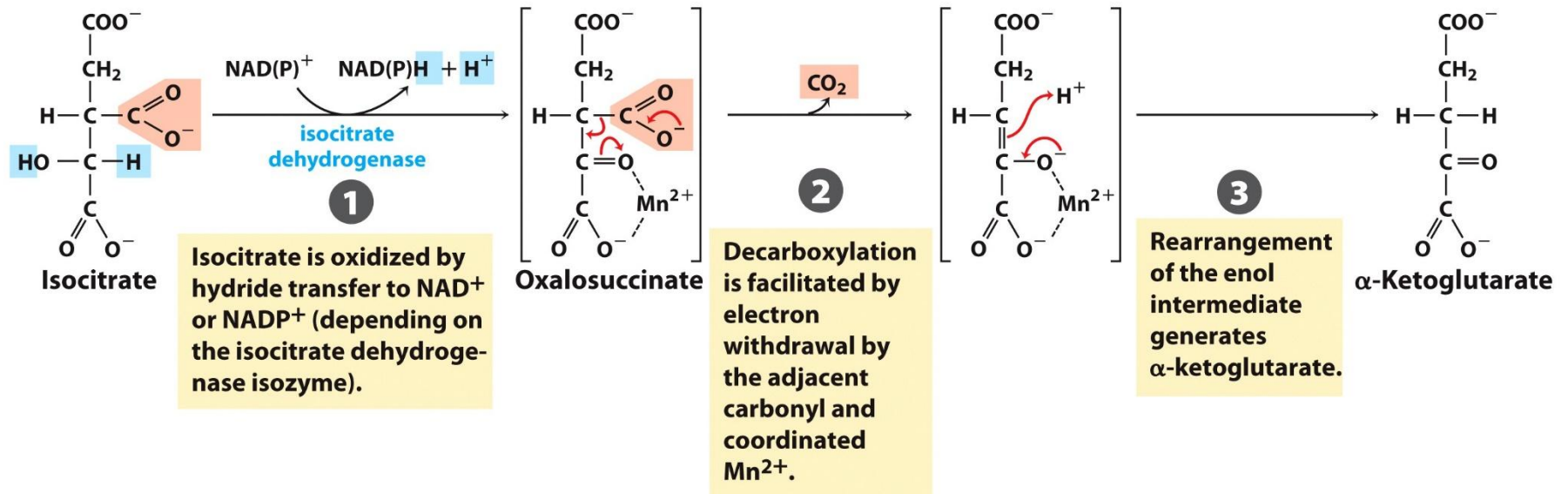
IRP bound to iron response element (IRE)?

Ferritin mRNA (iron storage) 5' IRE AAA(A)_n 3'	Ferritin mRNA translation		
	Ferritin synthesis		
	Repressed	Activated	
	Decreased	Increased	
Transferrin receptor (TfR) mRNA (iron transfer) 5' IREs AAA(A)_n 3'	TfR mRNA stability		
	TfR synthesis		
	Increased	Decreased	
	Increased	Decreased	

Iron homeostasis in our body



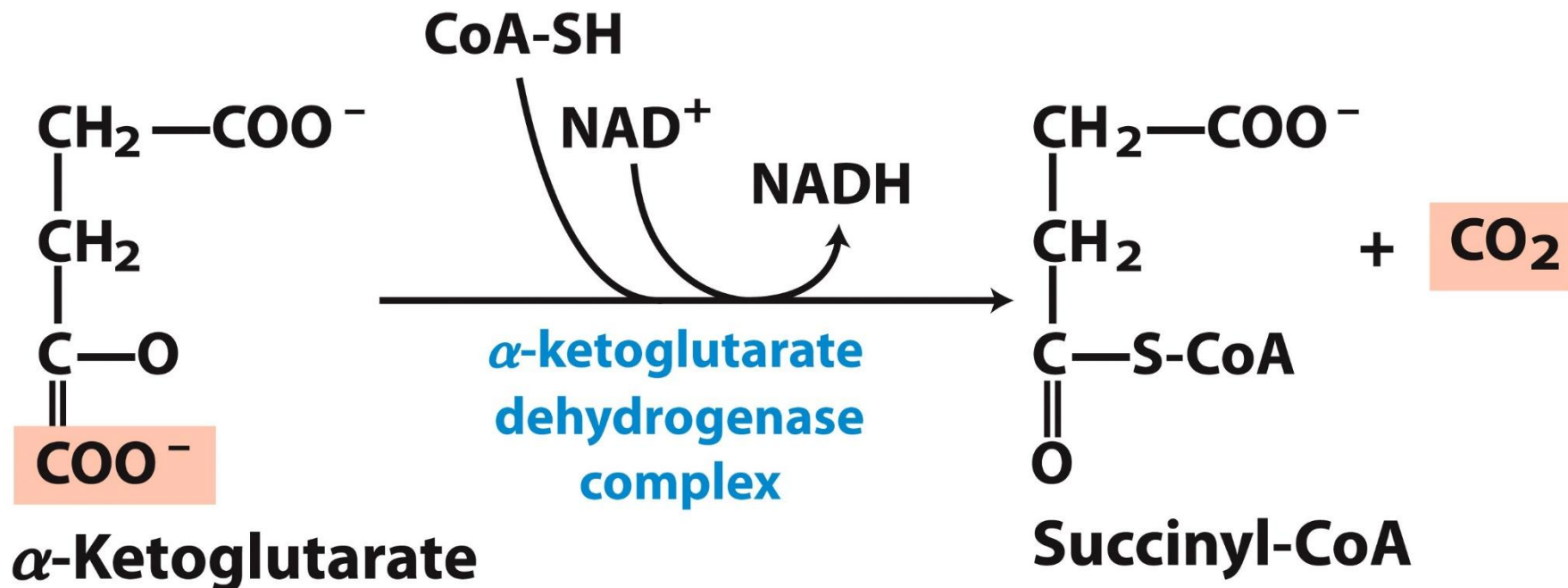
3) Isocitrate dehydrogenase



α -Hydroxyglutaric acid (2-hydroxyglutaric acid)

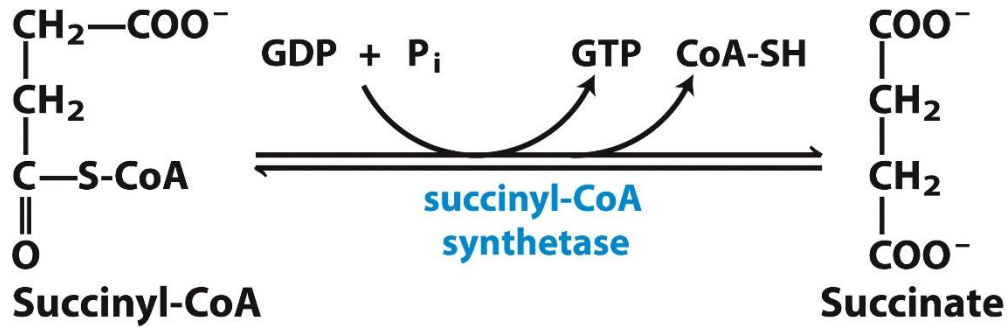
Mutations in **isocitrate dehydrogenase**, which frequently occur in glioma and AML, produce **D-2-hydroxyglutarate** from α -ketoglutarate. D-2-hydroxyglutarate inhibits the function of enzymes that are dependent on α -ketoglutarate, including **histone lysine demethylases**. $\rightarrow$ **cancer**

4) α -ketoglutarate dehydrogenase



$$\Delta G'^{\circ} = -33.5 \text{ kJ/mol}$$

5) succinyl-CoA synthetase



$$\Delta G'^{\circ} = -2.9 \text{ kJ/mol}$$

Unnumbered 16 p645

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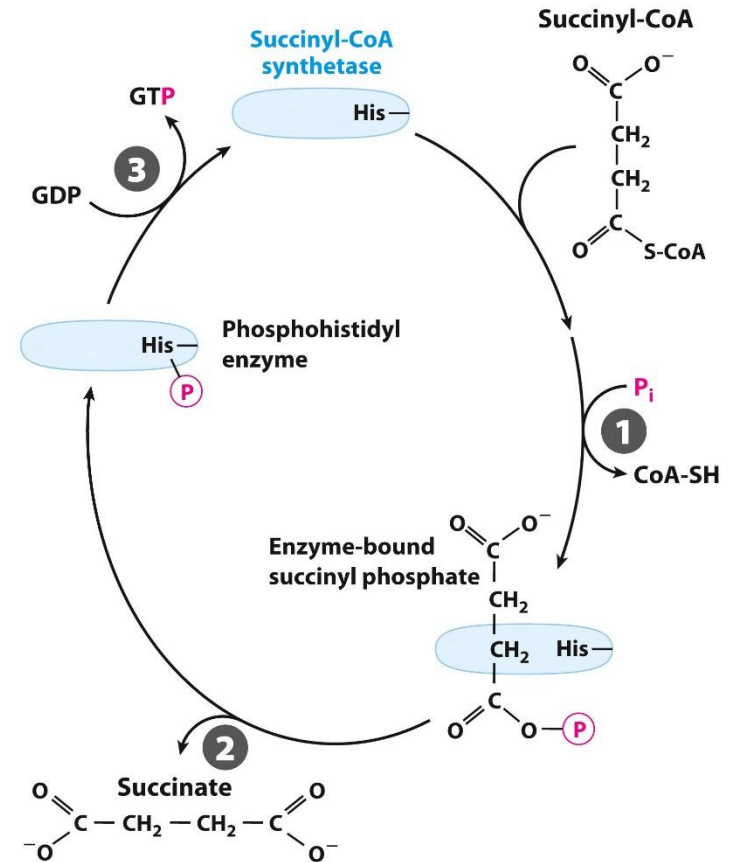
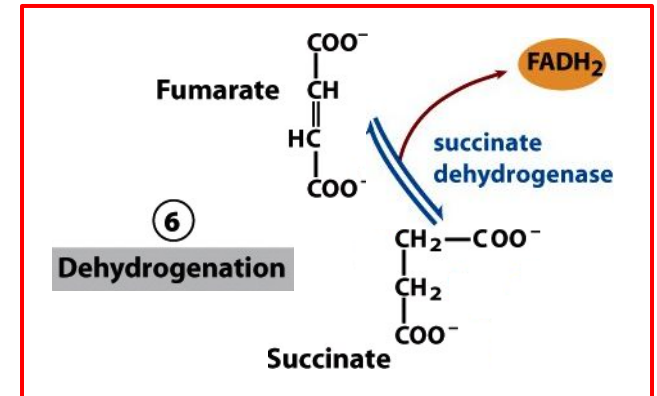
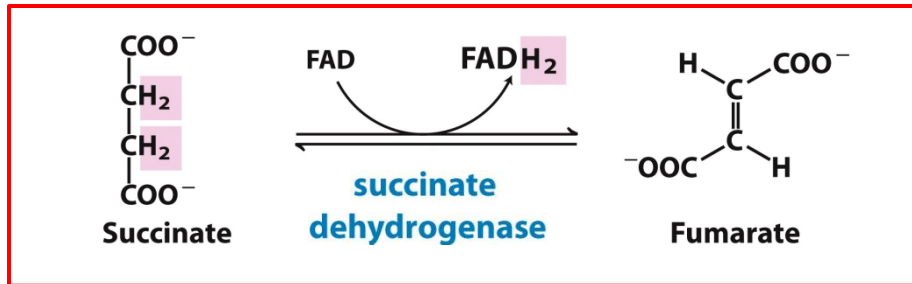


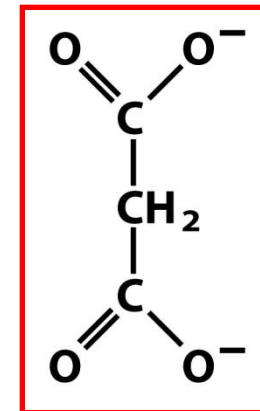
Figure 16-13a

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6) Oxidation of succinate to fumarate ($\Delta G'^0 = 0 \text{ kJ/mol}$)

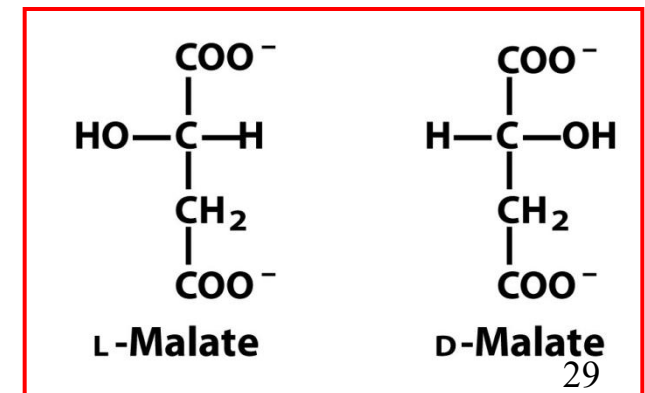
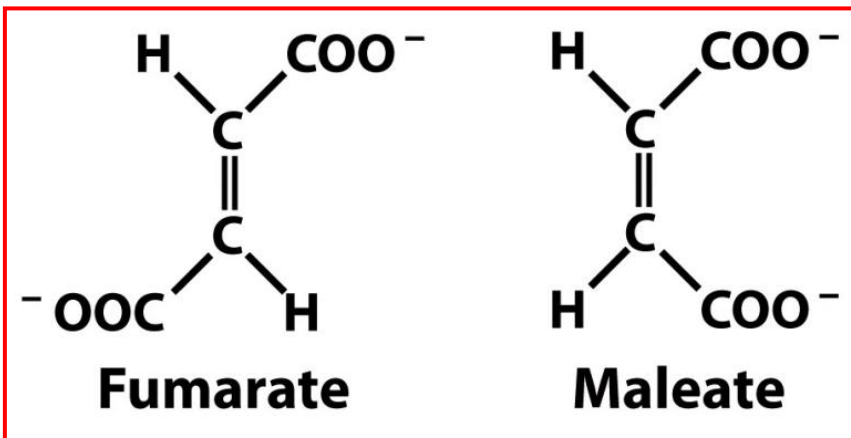
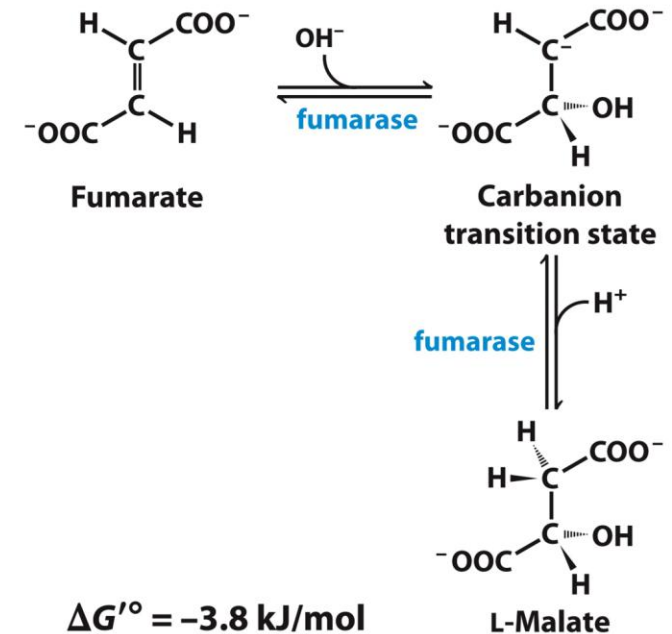


- **FADH₂** produced
- **Malonate is a strong competitor**
cycle can be blocked.

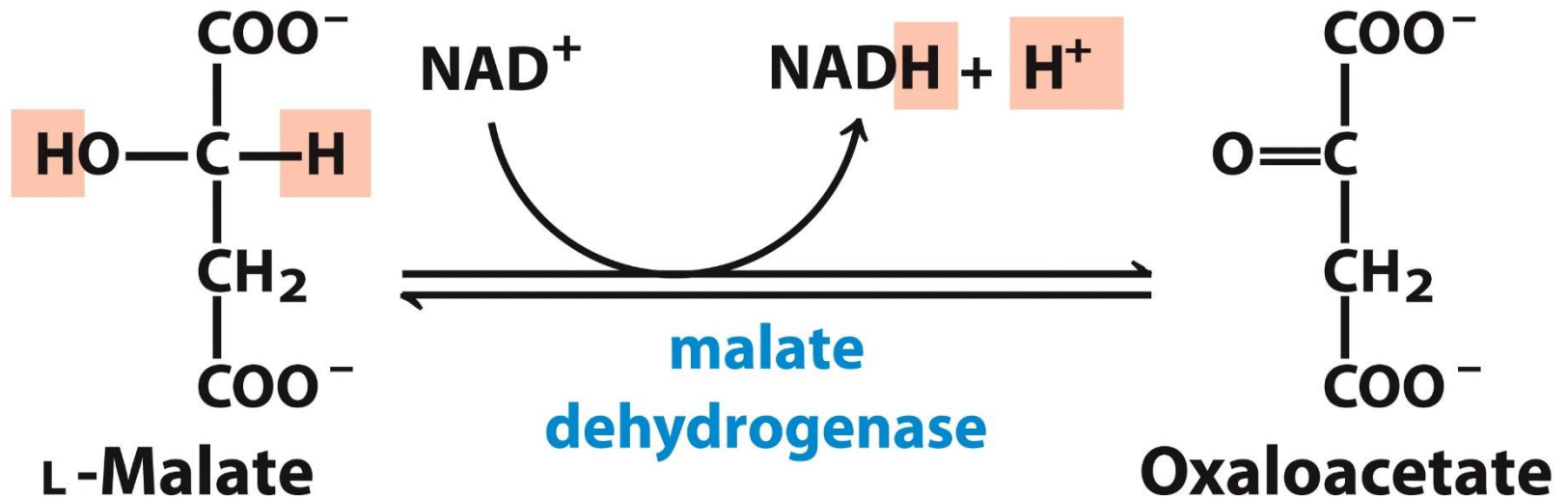


7) Hydration of fumarate to malate ($\Delta G'^{\circ} = -3.8 \text{ kJ/mol}$)

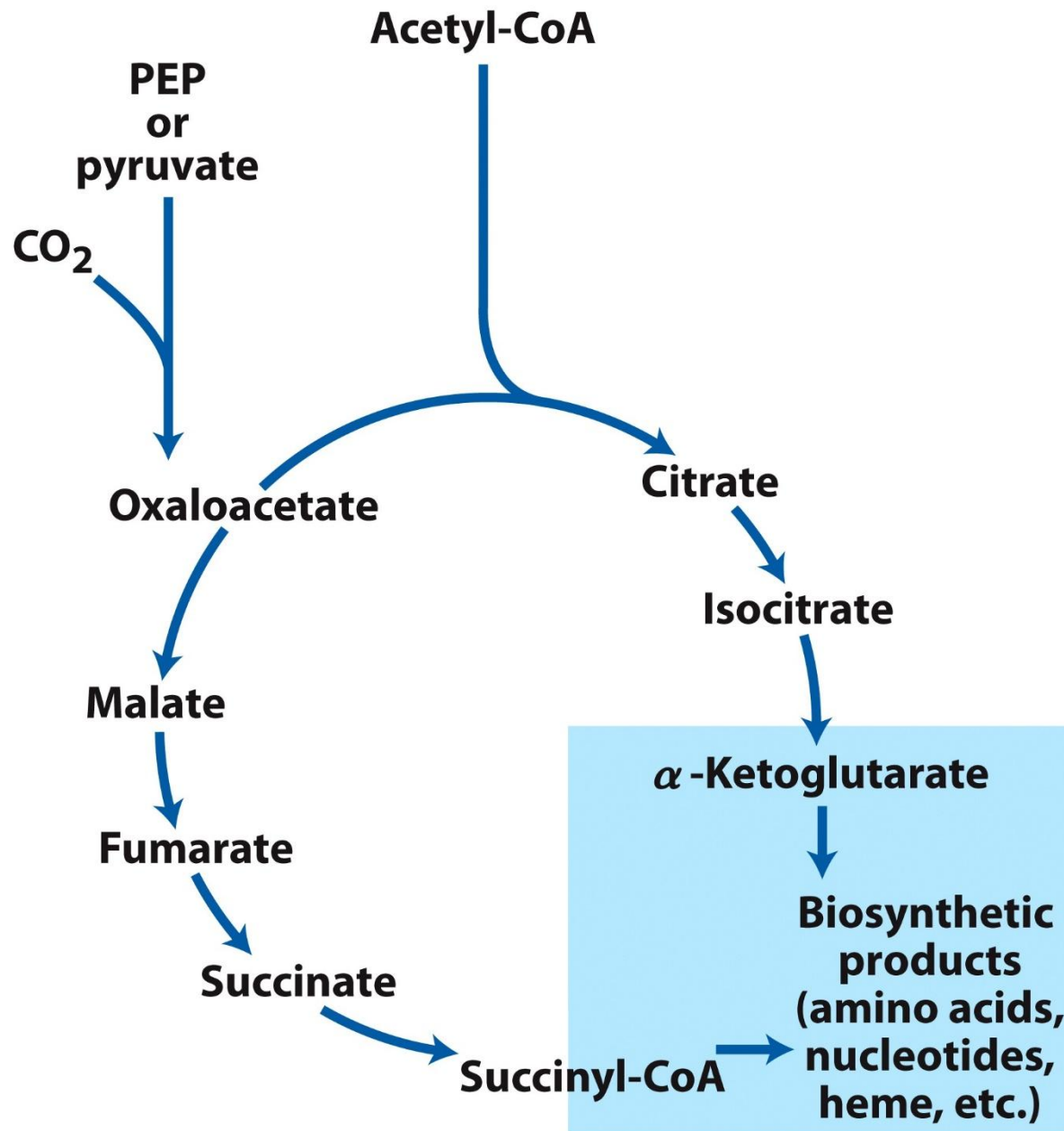
- Fumarase is **highly stereo-specific**: works on **trans double bond** of fumarate (not cis double bond of maleate)
- Reverse rxn: fumarase works on **L-malate** (not D-malate)



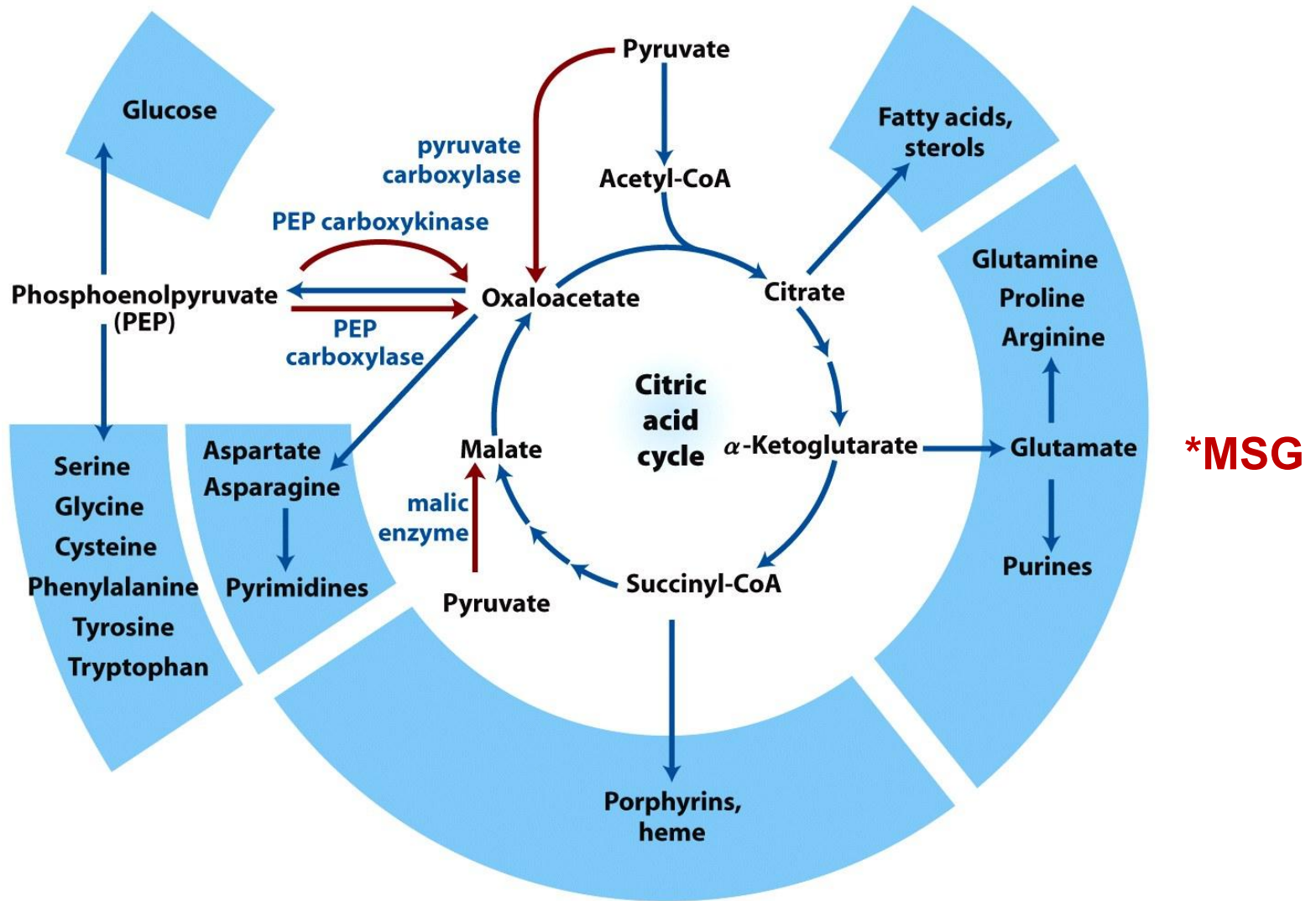
8) Malate to oxaloacetate ($\Delta G'^{\circ} = 29.7 \text{ kJ/mol}$)



$$\Delta G'^{\circ} = 29.7 \text{ kJ/mol}$$



Role of the citric acid cycle in anabolism



→ : anaplerotic reactions : 생합성을 위해 빠져나간 대사체 보충



잠시 쉬었다 가요...

Today, in this class...

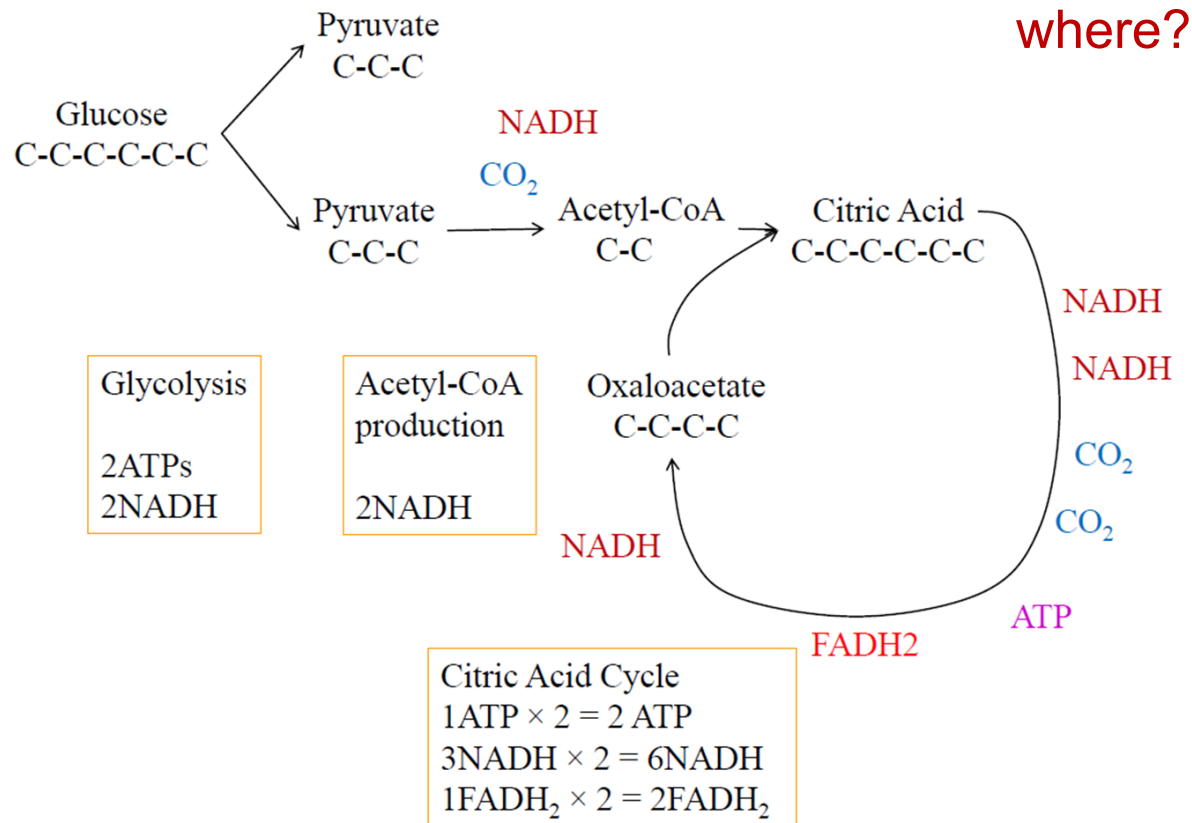


Regulation of citric acid cycle

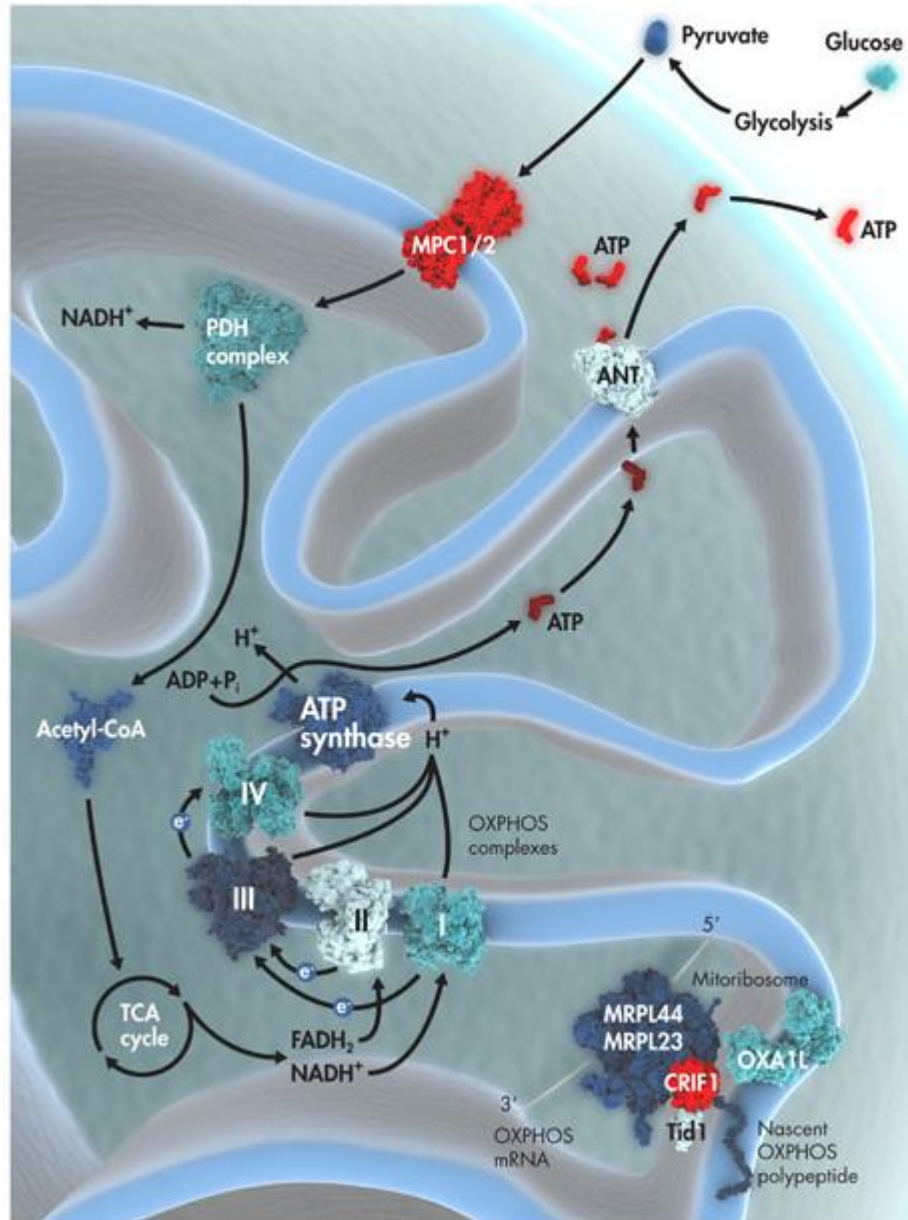


Glyoxylate cycle

16.3 Regulation of Citric Acid Cycle



Mitochondrial pyruvate carrier (MPC)

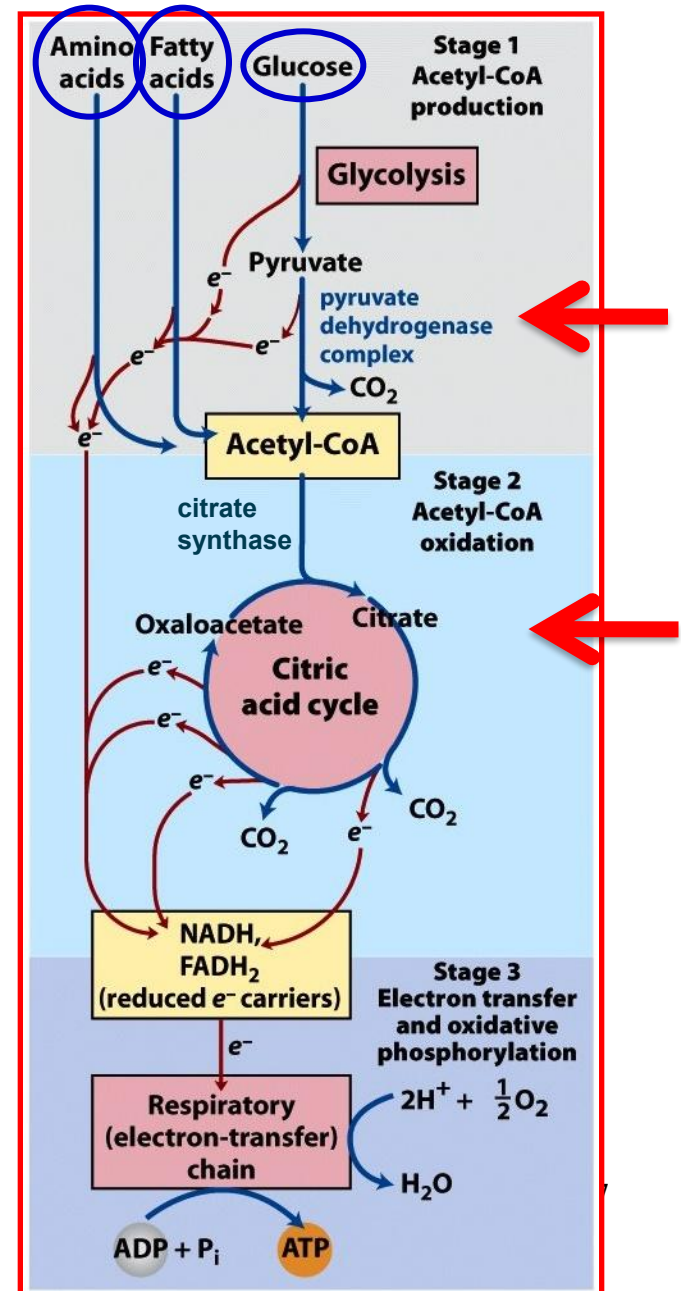


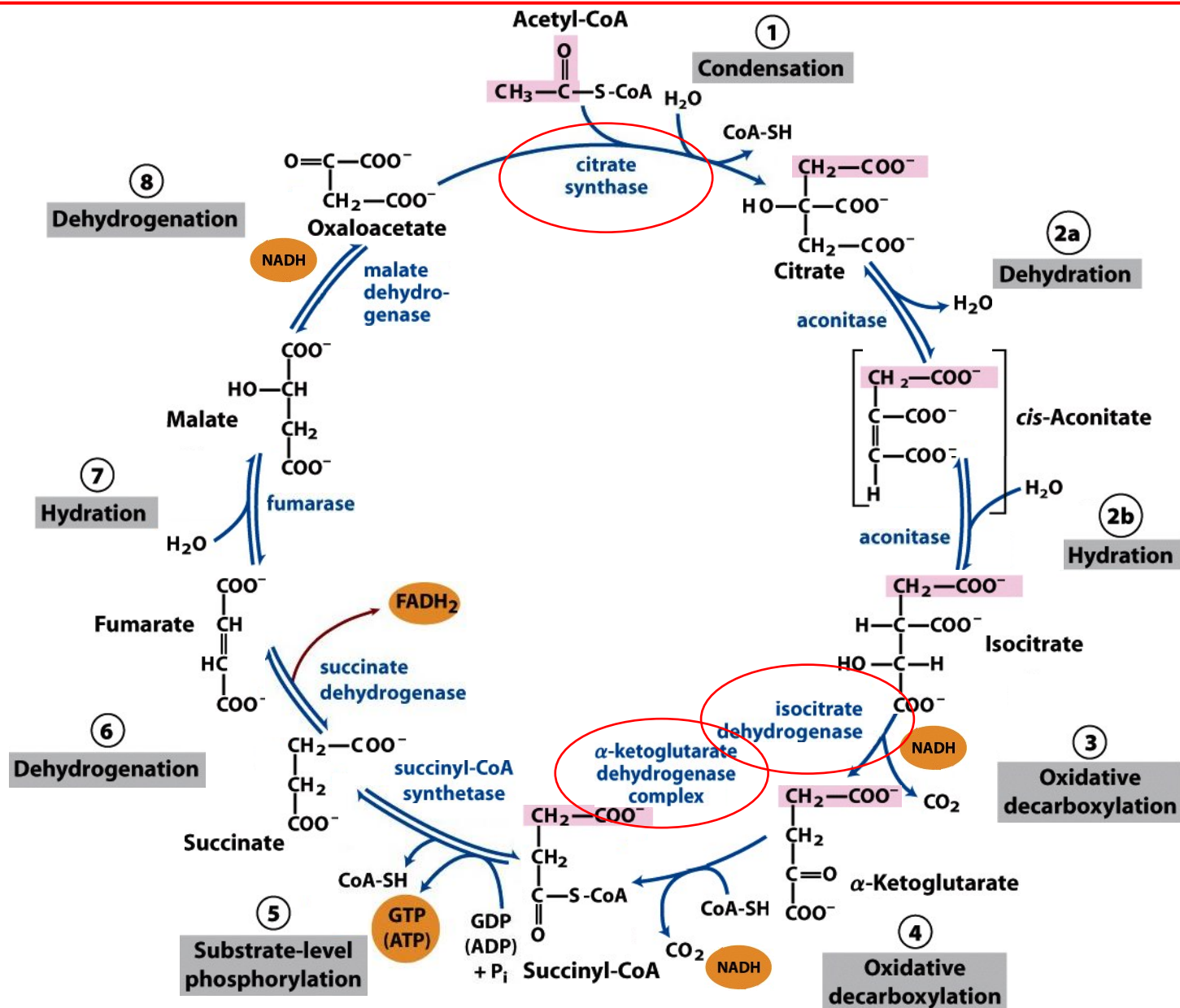
Adenine nucleotide translocator (ANT)

Mitochondrial ribosome complex

Regulation of Citric Acid Cycle

- Cycle is tightly regulated at pyruvate DH step & 3 exergonic reactions (citrate synthase step, isocitrate DH step, a-ketoglutarate DH step)
- Sources of Acetyl CoA
 - Glucose (thru pyruvate)
 - Fatty acids
 - Amino acids

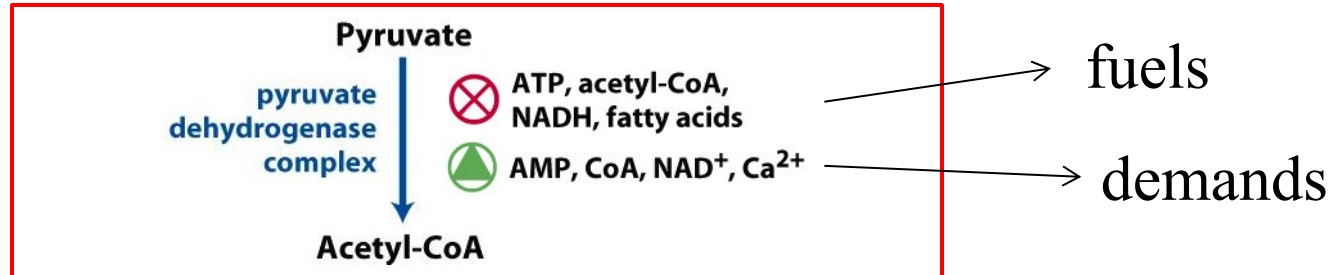




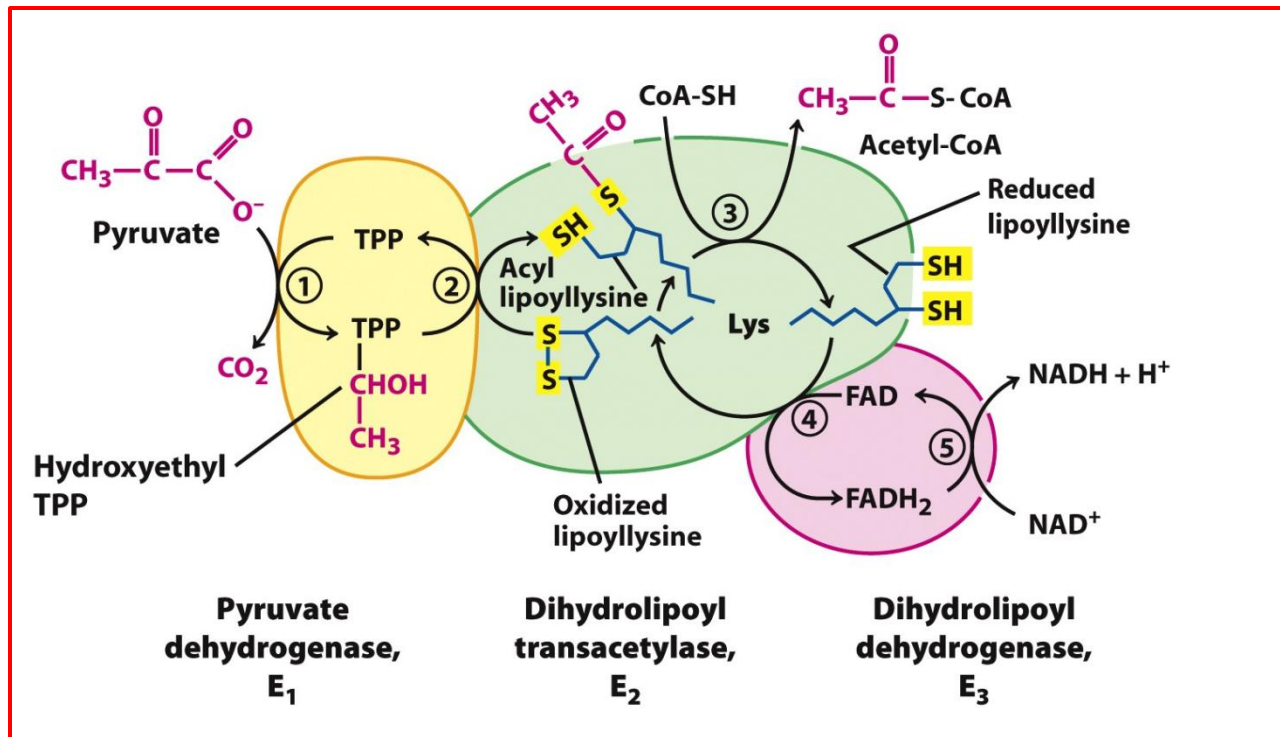
3 exergonic reactions

Regulation of Citric Acid Cycle

1) Allosteric and covalent mechanisms on PDH

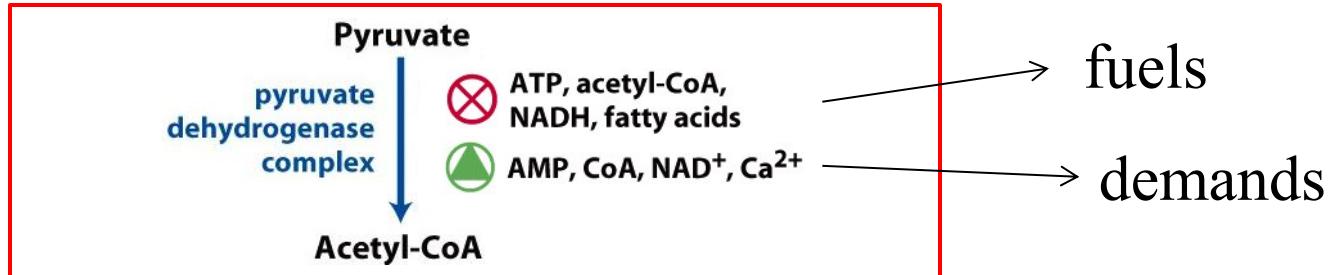


– PDH is inhibited by phosphorylation of E1

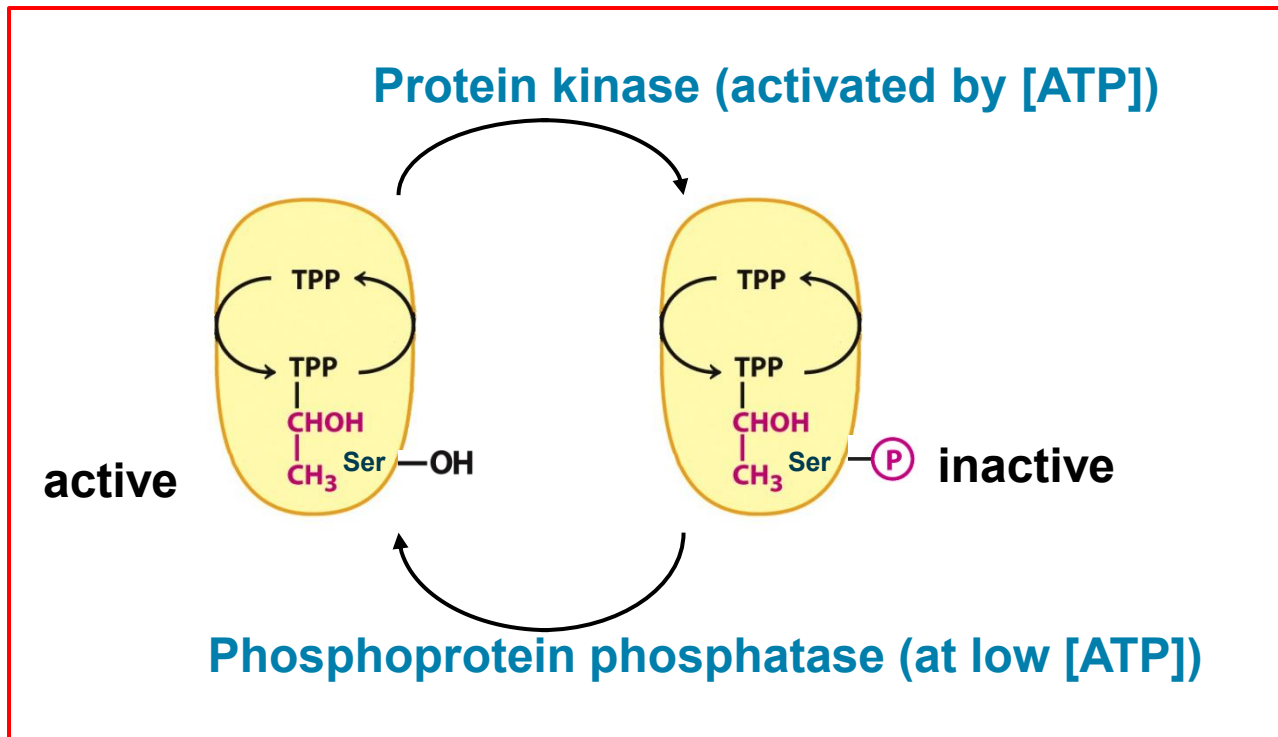


Regulation of Citric Acid Cycle

1) Allosteric and covalent mechanisms on PDH

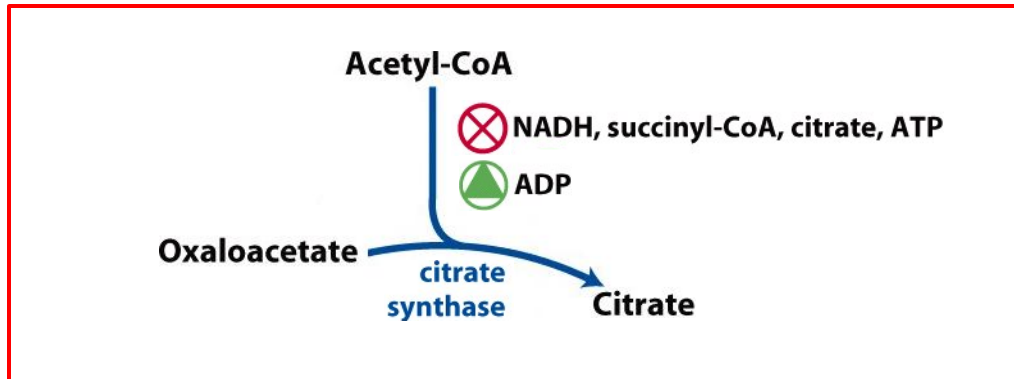


– PDH is inhibited by phosphorylation of E1



2) Regulation at 3 exergonic reactions

① Citrate synthase



Substrate availability (**acetyl CoA, OAA**) can limit the reaction.

Product accumulation (**citrate**) inhibits citrate synthase

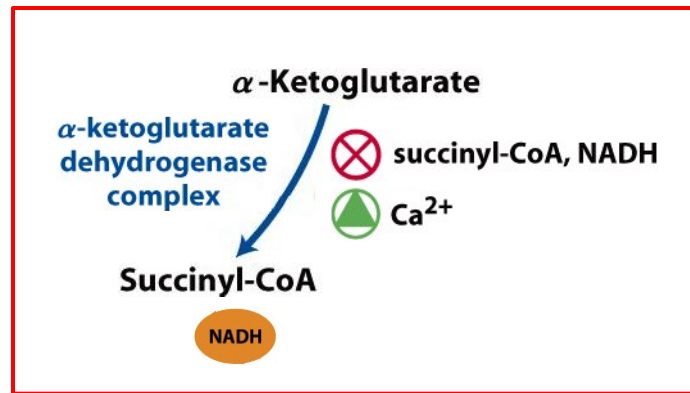
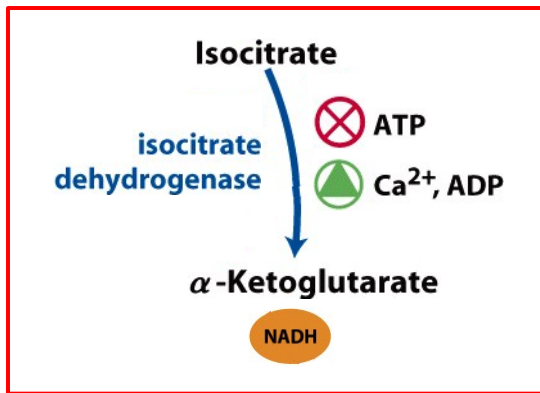
Succinyl-CoA, ATP, NADH inhibits citrate synthase

(’22 약시)

2) Regulation at 3 exergonic reactions

② Isocitrate dehydrogenase

③ α -kg dehydrogenase



both inhibited by high energy : **ATP, NADH**

- These are limited by high **[Product]** ('22 약시)
- They are stimulated by **Ca^{2+}**

Citric acid cycle is regulated by :

$[ADP] / [ATP]$

$[NAD^+] / [NADH]$

$[substrates] / [products]$

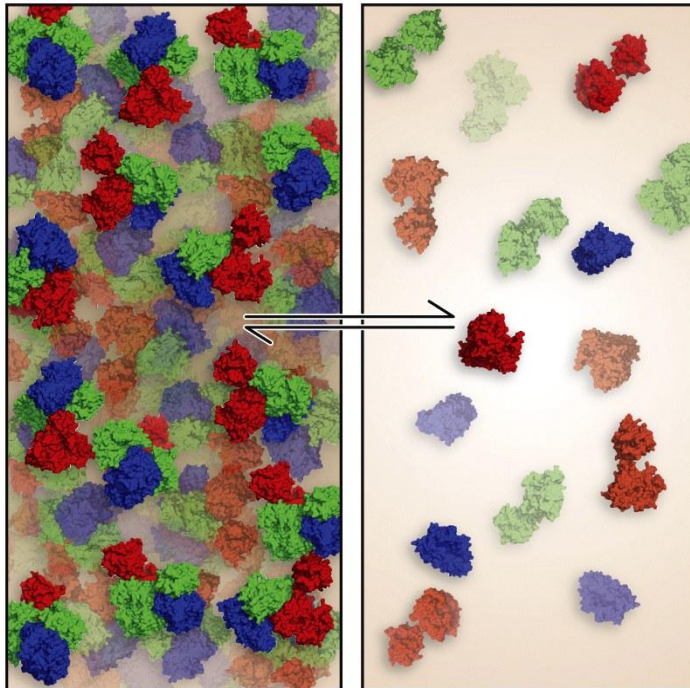
$[demands] / [fuels]$

$[Ca^{2+}]$ activates PDH, IDH and KDH

Why $[Ca^{2+}]$ affects the citric acid cycle?

3) **Substrate channeling** thru multi-enzyme complexes

- Enz's of citric acid cycle are soluble components
 - Enz's exist as multienzyme complex within mitochondria
 - Enz's are highly organized in complexes
- Channeling can make a metabolic pathway more rapid and efficient. It prevents the release of unstable intermediates.



High concentration favors association

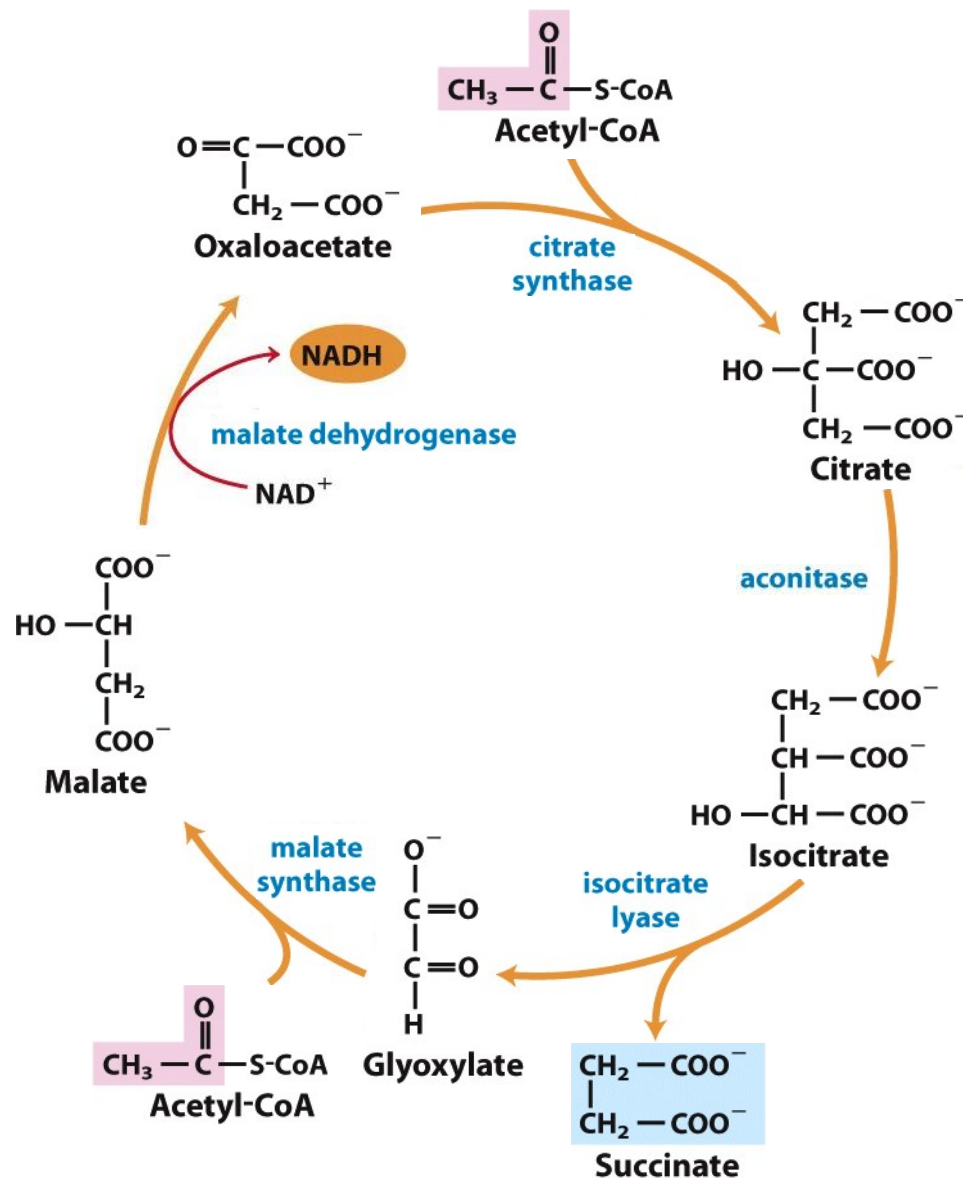
The inside of the cell is crowded.
Mitochondria are even more so.



What if the cell is disrupted?

16.4 The Glyoxylate Cycle

The Glyoxylate Cycle



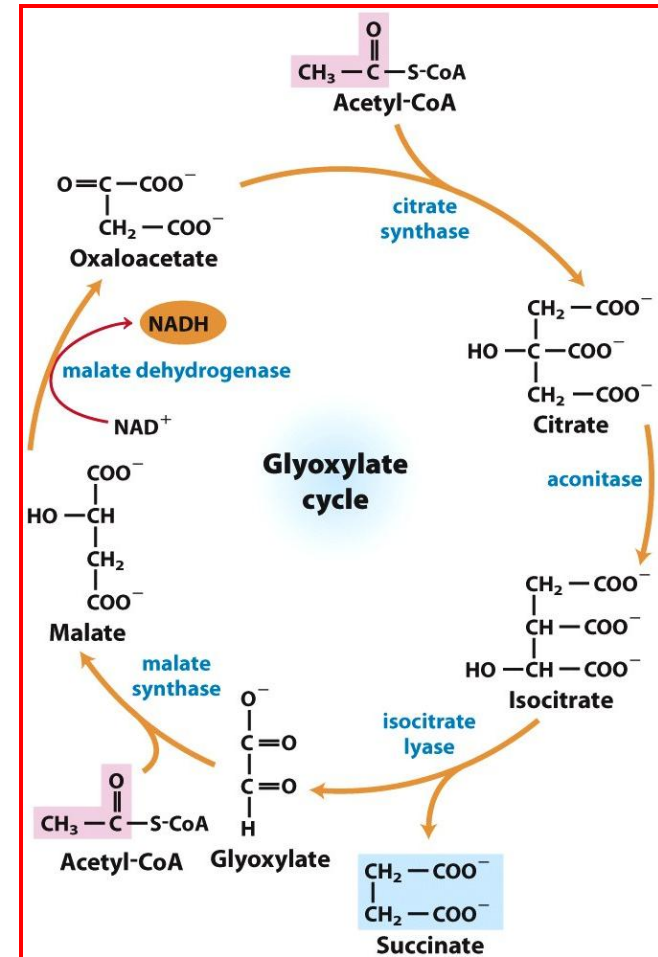
Input ?

Outcome ?

Difference from
TCA cycle ?

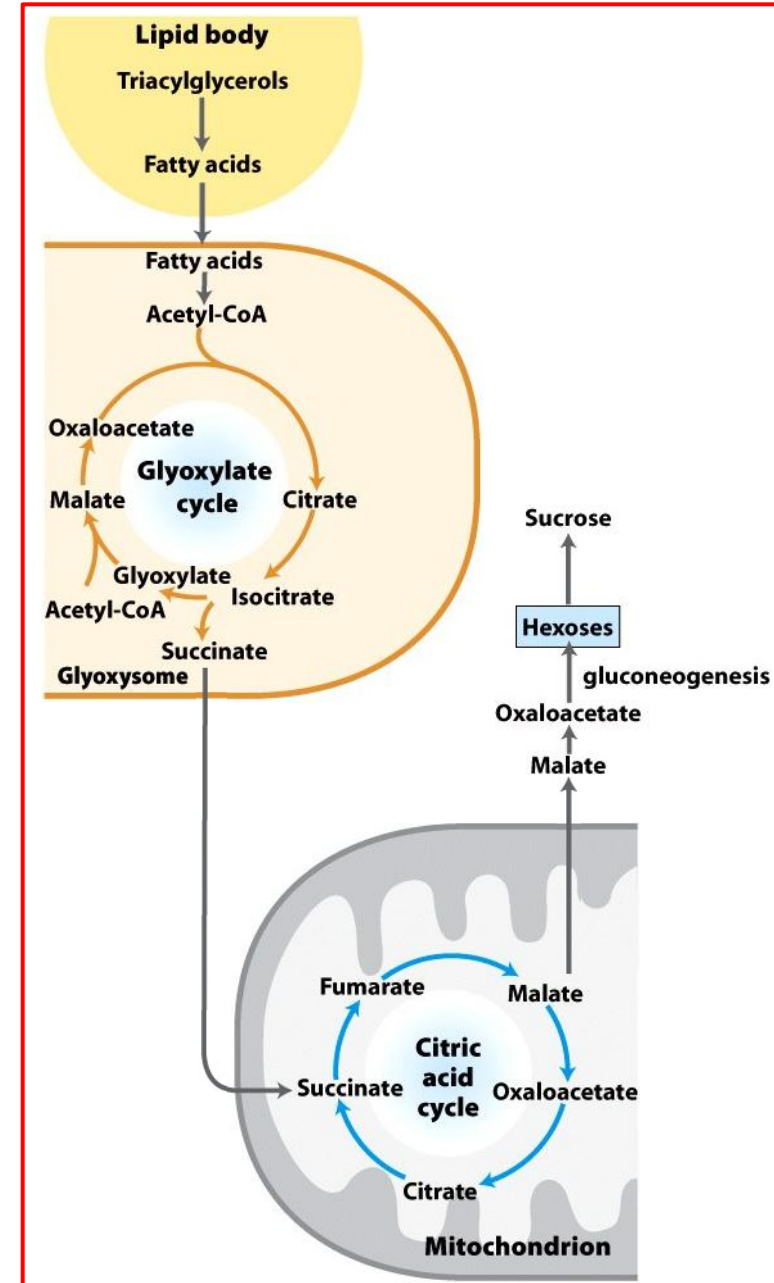
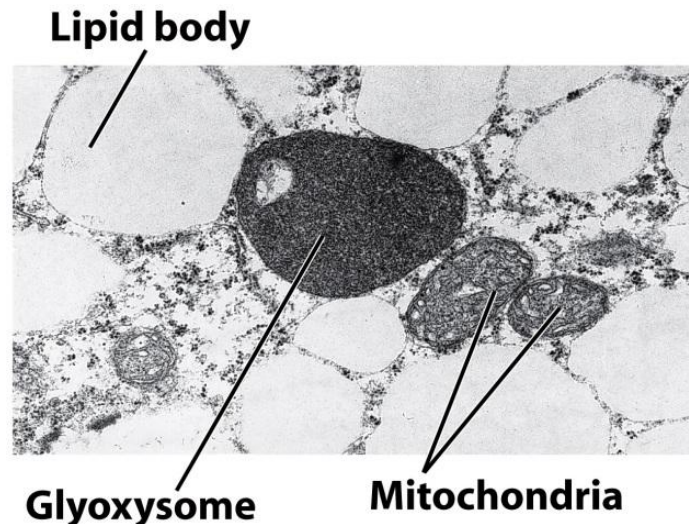
1) 4-C compounds (succinate) from 2 acetates

- Isocitrate cleaved into glyoxylate and succinate by isocitrate lyase
 - Glyoxylate + acetyl CoA → malate → oxaloacetate (Glyoxylate cycle)
 - **Succinate** → fumarate → malate (TCA cycle) ... → gluconeogenesis
 - Glyoxysome in lipid-rich germ
 - Degradation of FA to acetyl CoA
 - Acetyl CoA to succinate
 - Gluconeogenesis
- Lipids → Glucose.**



2) Citric acid & glyoxylate cycles are connected through succinate

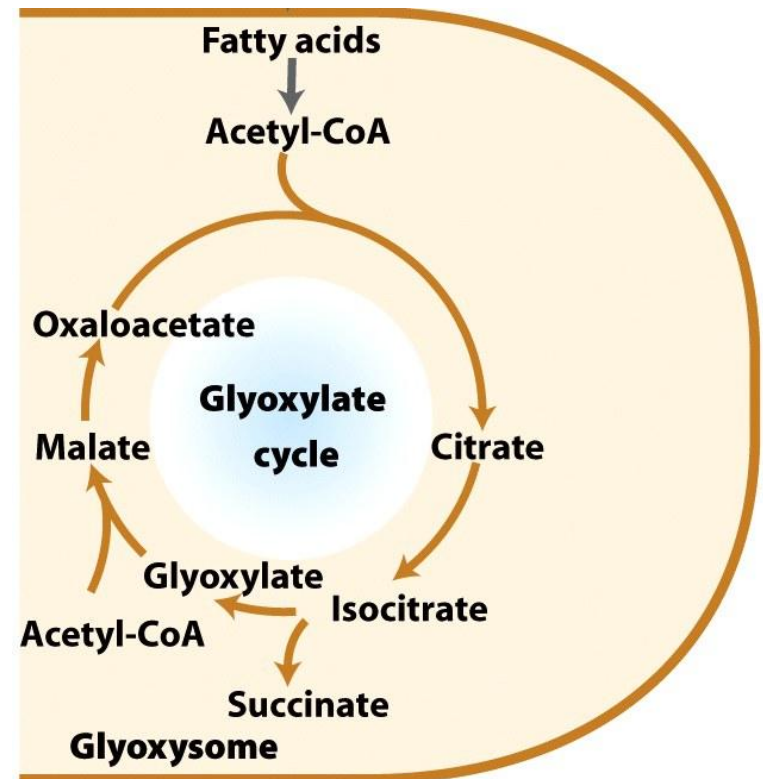
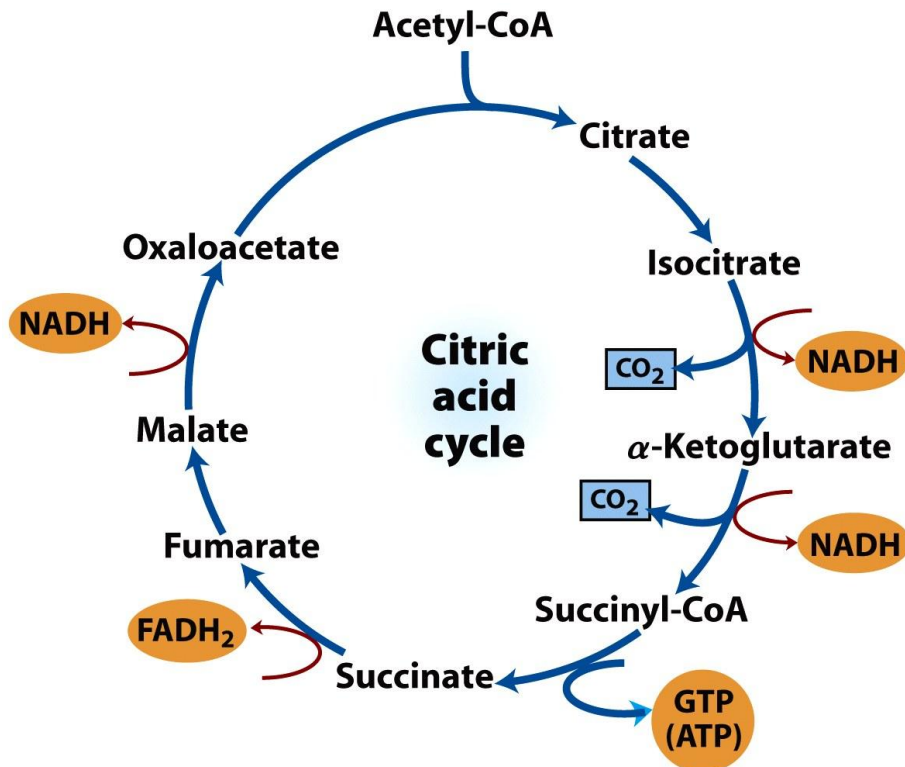
- Succinate is transported into mitochondria for citrate cycle
- Malate carried to cytosol (gluconeogenesis)
- To roots and shoots



Vertebrates cannot convert fatty acids to carbohydrates ... why ?

Acetyl-CoA $\rightarrow$ 2 CO₂

Fatty acids $\rightarrow$ acetyl-CoA
 $\rightarrow$ Succinate $\rightarrow$ Malate $\rightarrow$
Oxaloacetate $\rightarrow$ PEP $\rightarrow$ G



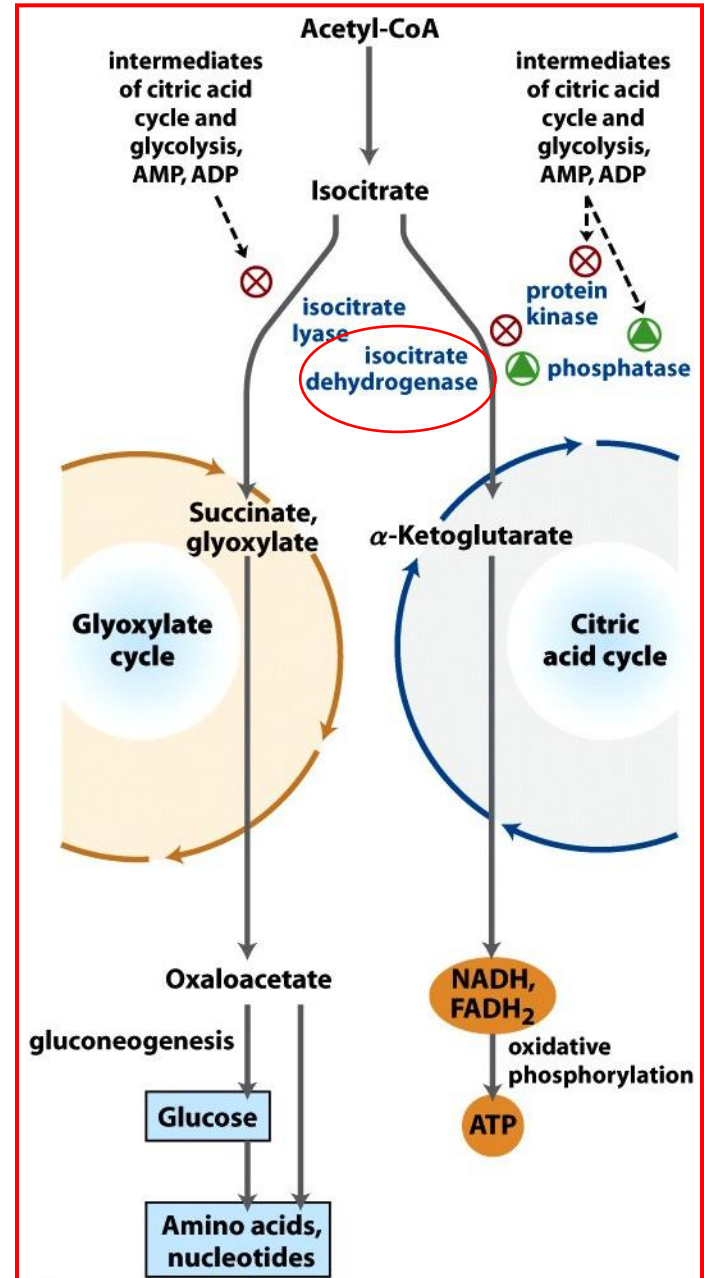
3) Regulation of the cycles

- Isocitrate dehydrogenase is regulated by phosphorylation
- Protein kinase: inactivates it
→ more to glyoxylate cycle
- PPase: remove P, activates it
→ more to citric acid cycle

Intermediates of citrate cycle:

Low intermediates → glyoxylate cycle

High intermediates → citric acid cycle



Summary

Regulation of citric acid cycle:

By the conversion of pyruvate to acetyl-CoA

By the flux thru citrate synthase, isocitrate DH, α -KG DH

The Glyoxylate Cycle

Germinating seeds and certain microorganisms

Live on acetate as the sole carbon source

Succinate and glyoxylate for gluconeogenesis

Reversible phosphorylation regulates the cycles

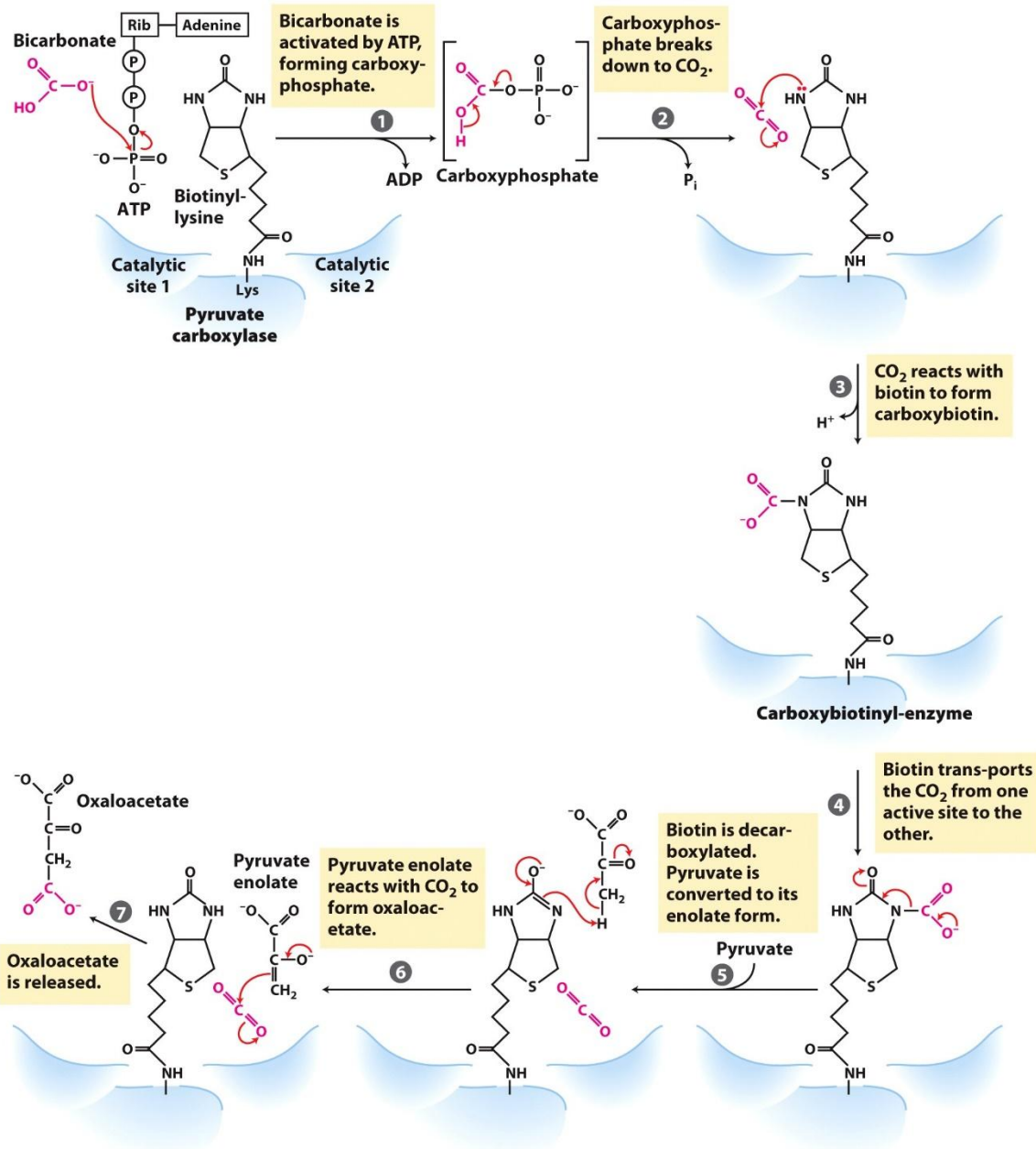


Figure 16-17

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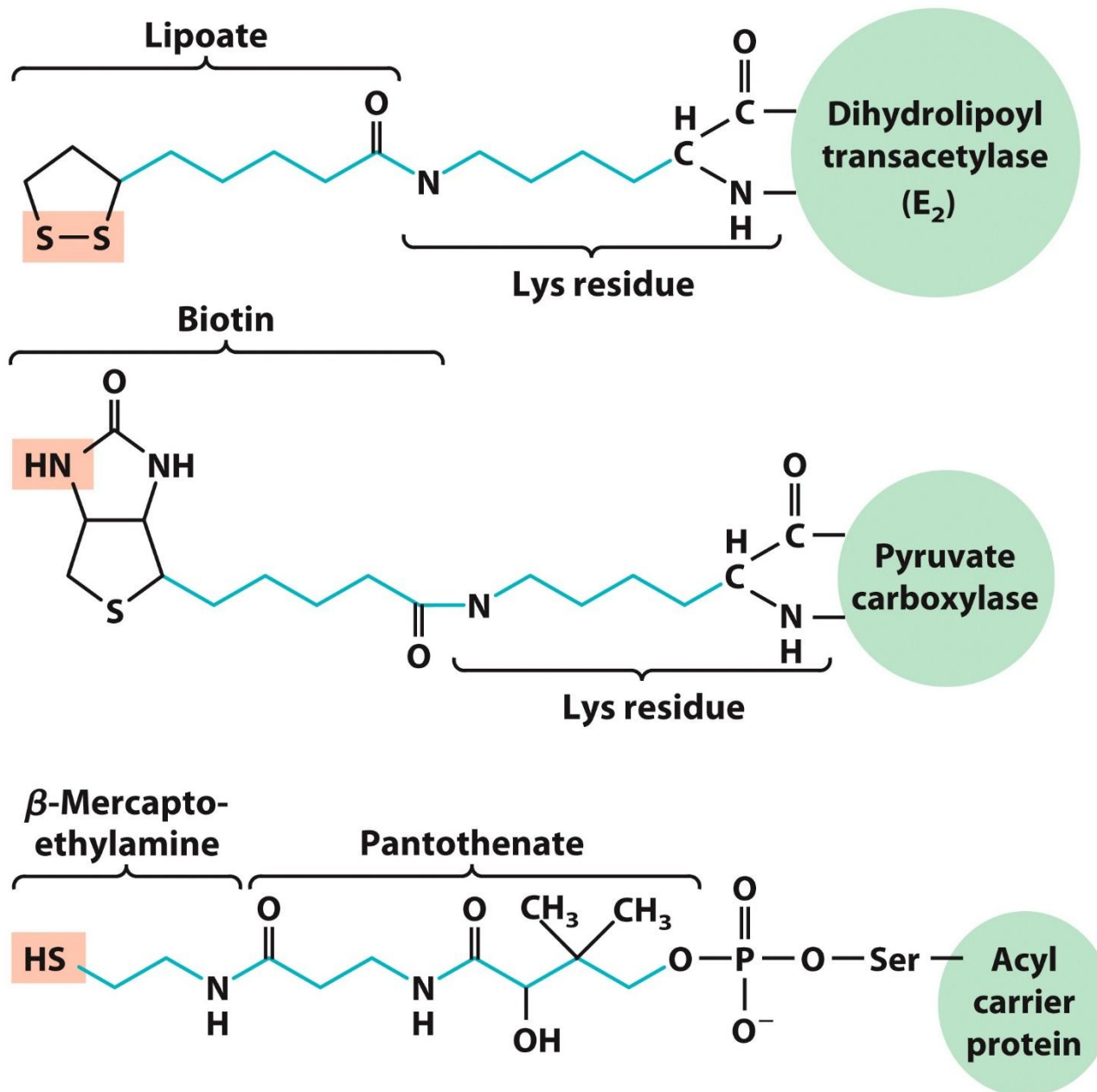


Figure 16-18

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